

Formation of complex optical fields with cascaded-quasiperiodic self-similar structures by multi-beam interference

以多光束干涉技術

生成具有級聯似週期自我相似性結構之複雜光場

May.28.2025

專題生：劉恆睿

指導教授：段必輝 博士

Outline

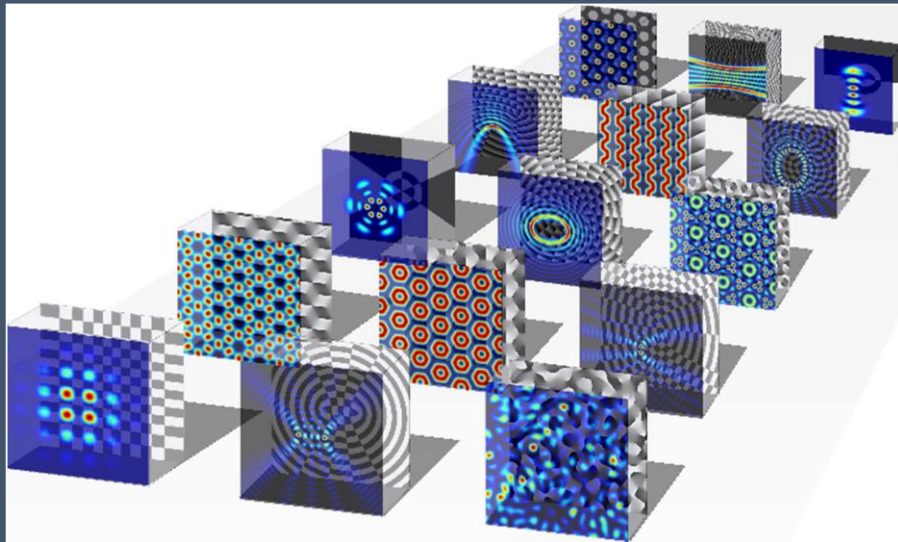
- Introduction
 - Background & Motivation
 - Crystal & Quasi-crystal
 - Self-similar structure
- Experimental principles & setup
- Photomask: configurations & fabrication results
- Experimental results & discussion
- Conclusion
- Future work
- Apology



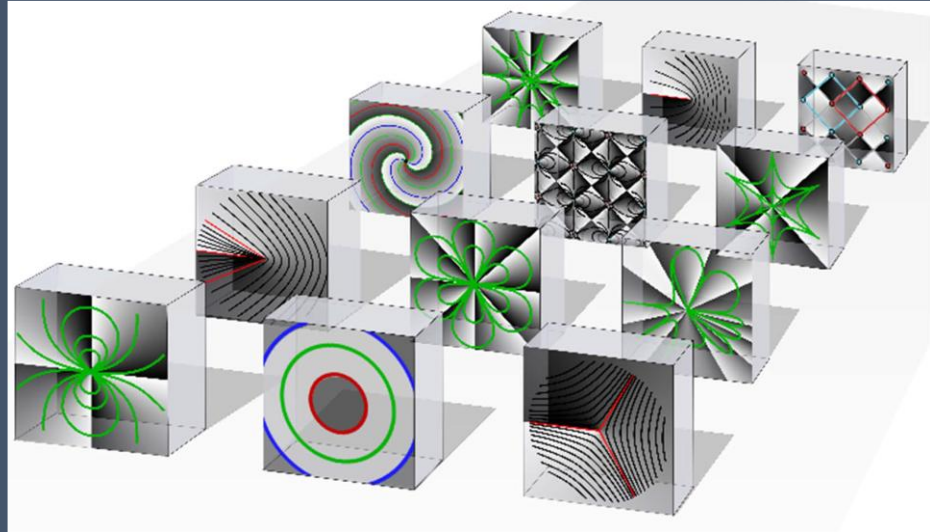
Structured light

Structured light means to tailor optical fields with customized **amplitude**, **phase**, and **polarization** distributions.

(a) Different **intensity** distribution structures



(b) Different **polarization** singular structures



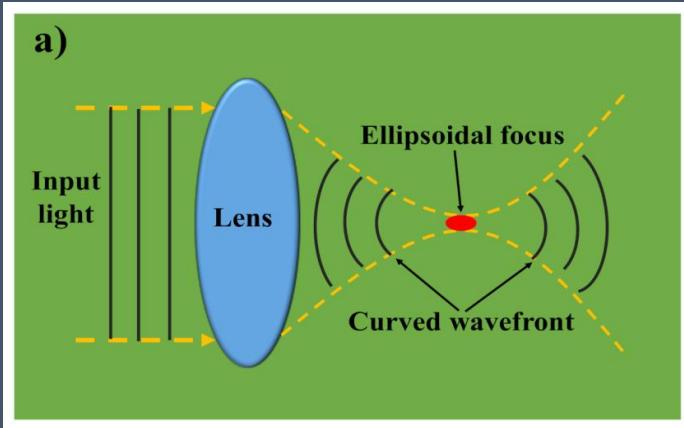
J. Opt. 19, 013001 (2017).

A key structured light: Bessel beam

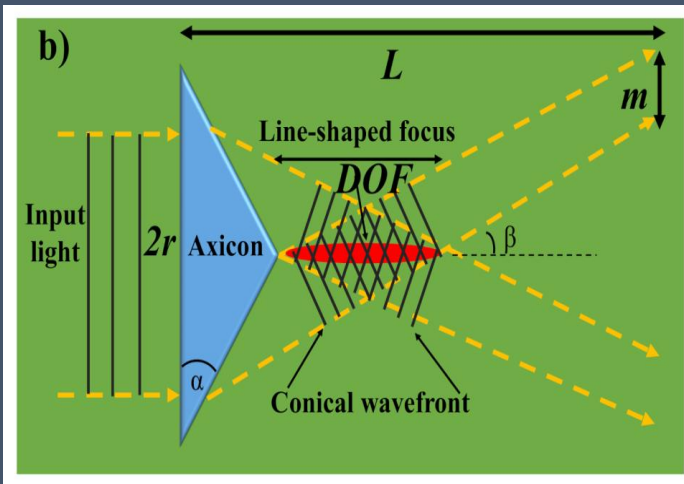


Bessel beam

Gaussian beam

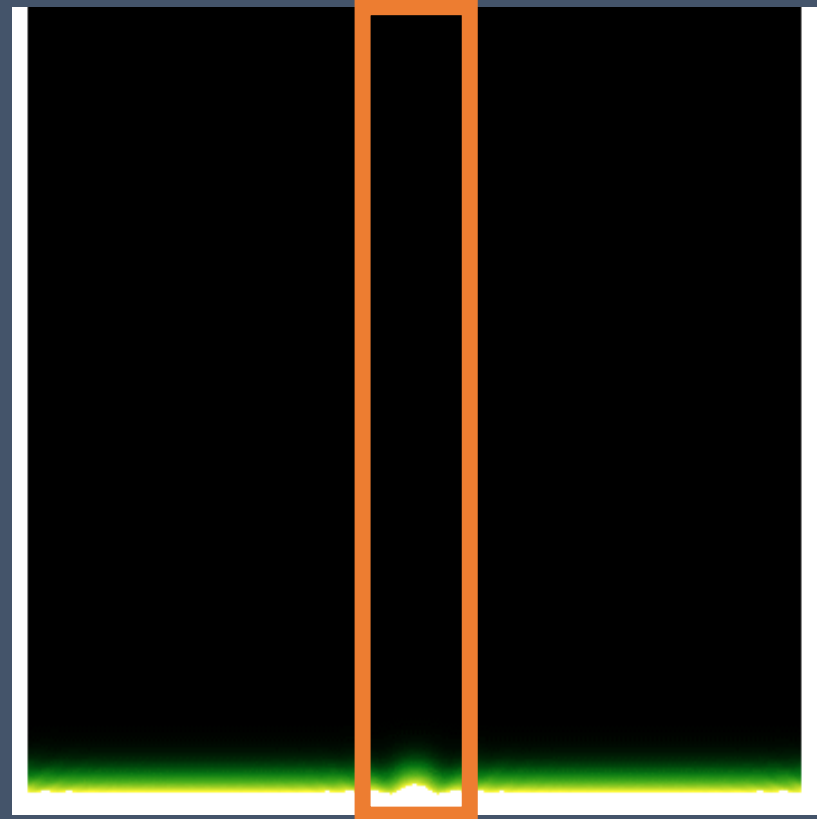


Bessel beam



Micromachines 11, 997 (2020)

Non-diffracting feature

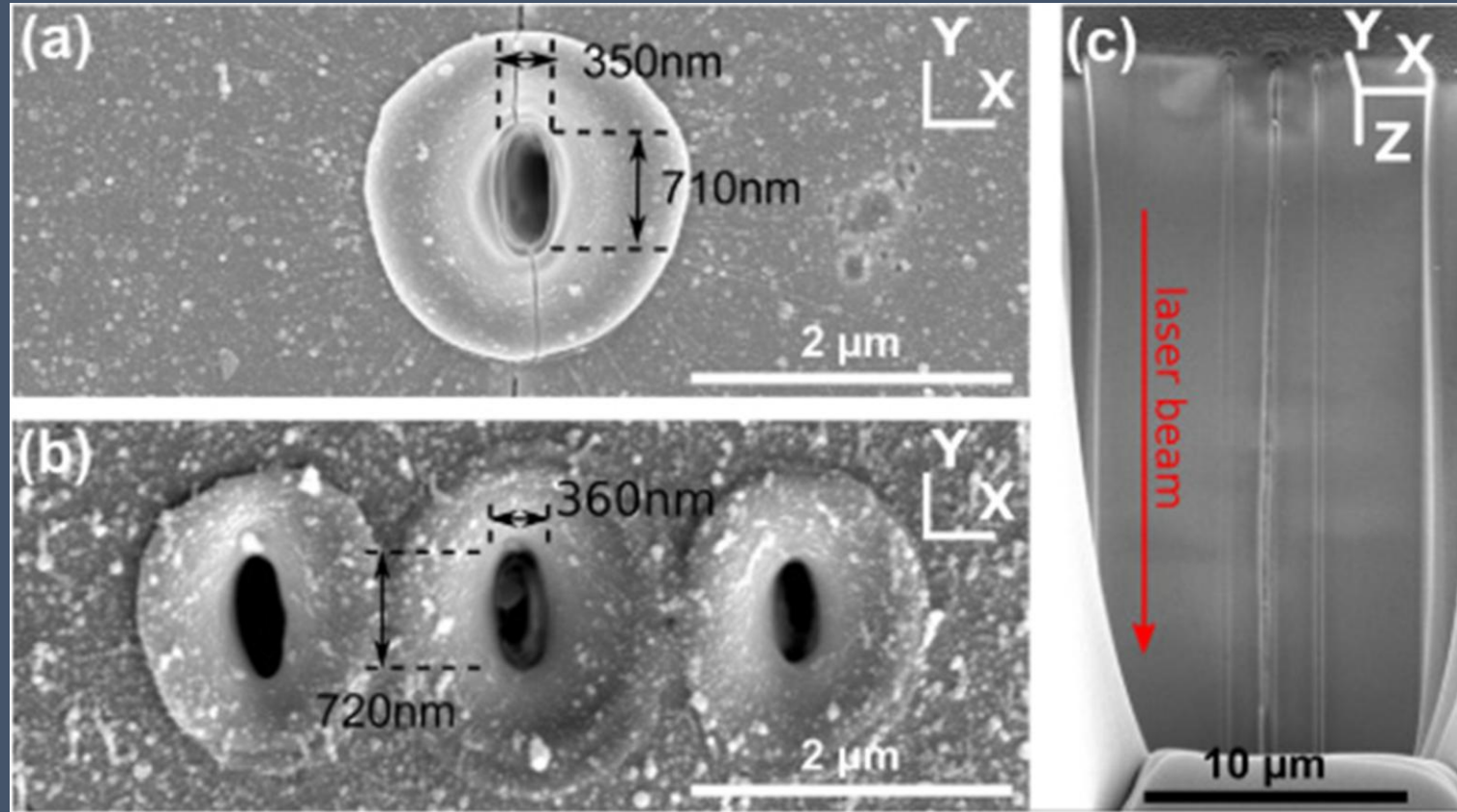


https://x.com/j_bertolotti/status/1059037440810323974



Application of Bessel beam

Laser engraving

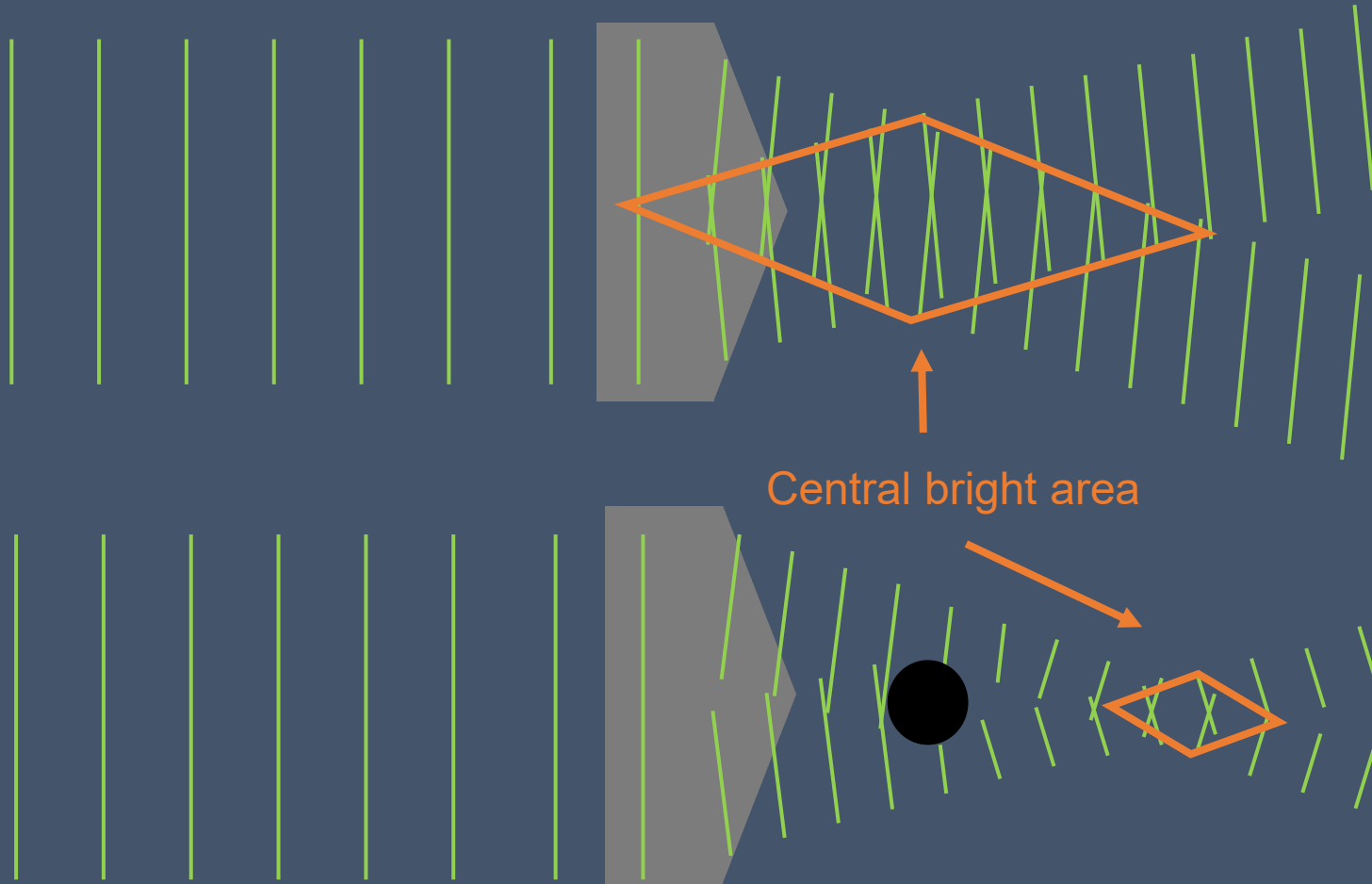


Opt. Lett. 42, 4307-4310, 2017



Bessel beam

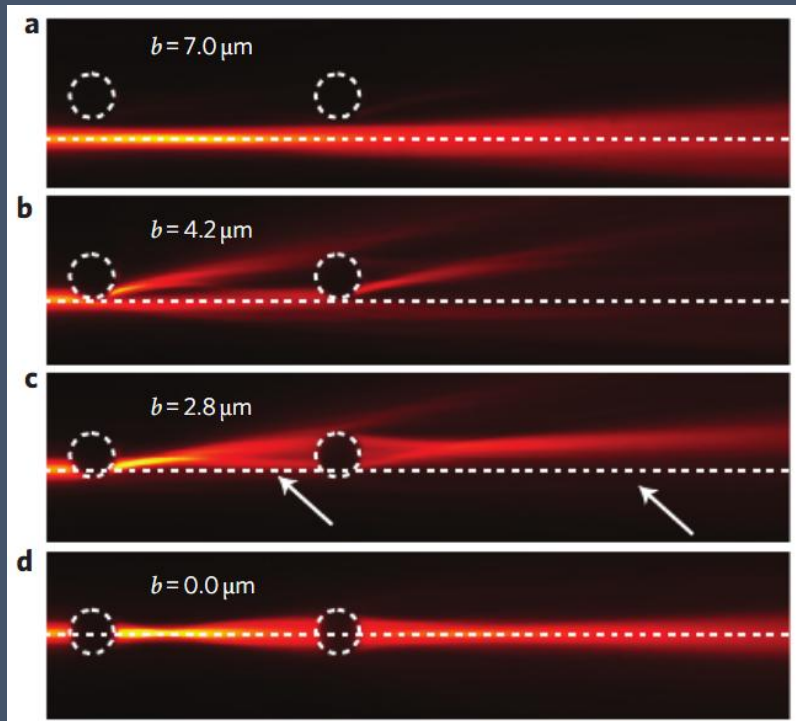
Self-healing feature



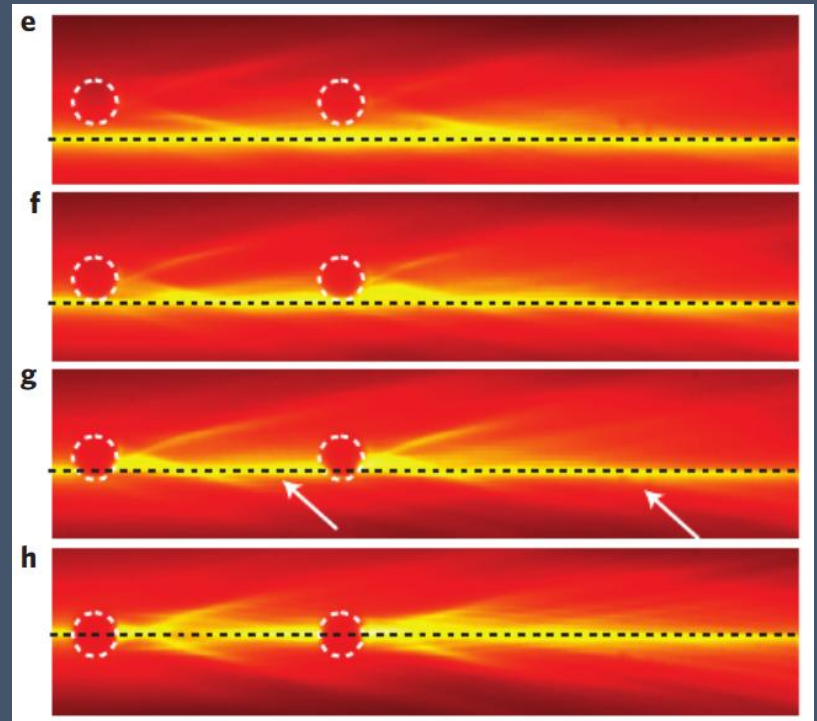
Application of Bessel beam

Fluorescence microscopy

Gaussian beam



Bessel beam



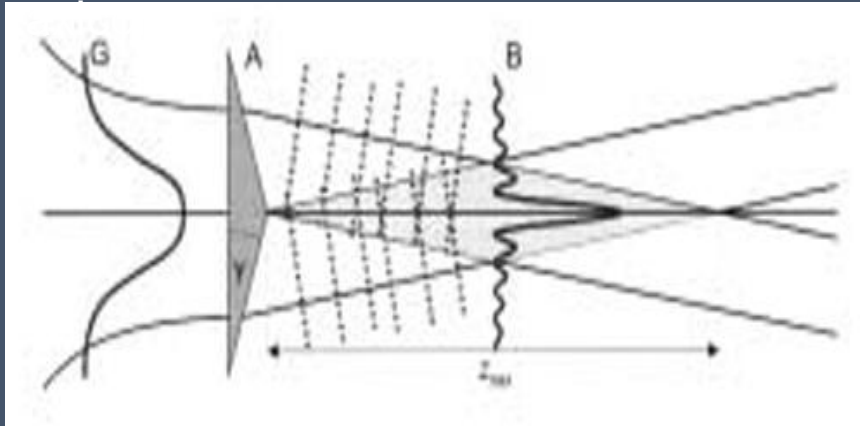
Nat. Photonics 4, 780 (2010)



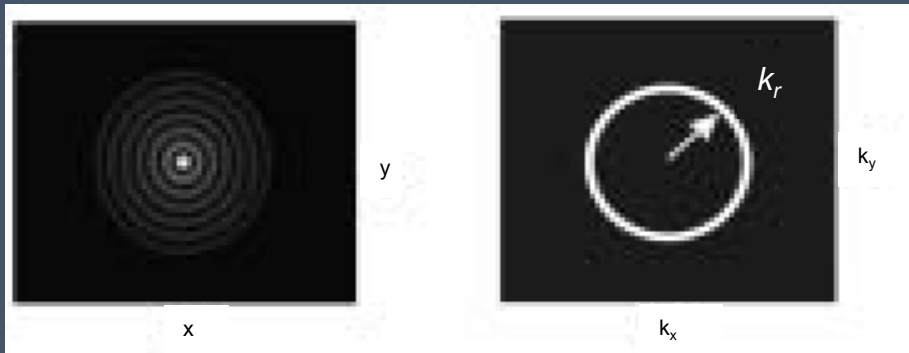
How to Create Bessel beams

Low cost way:

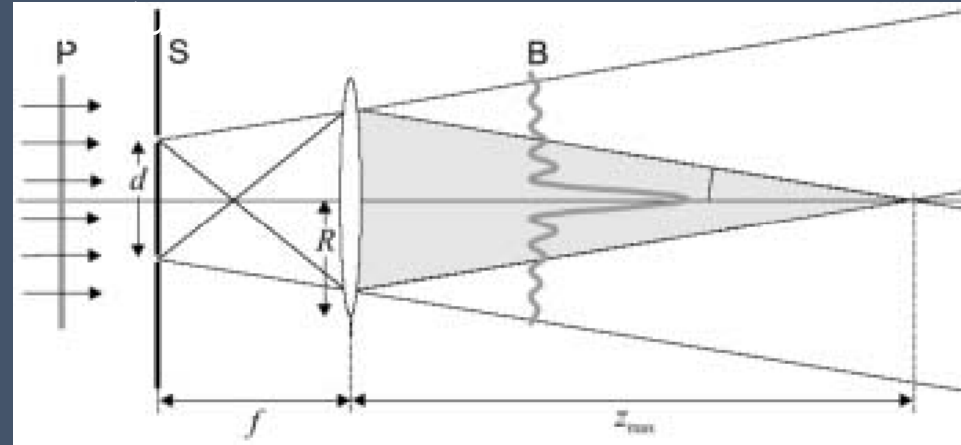
① Formed by



Intensity profile



② Formed by annular



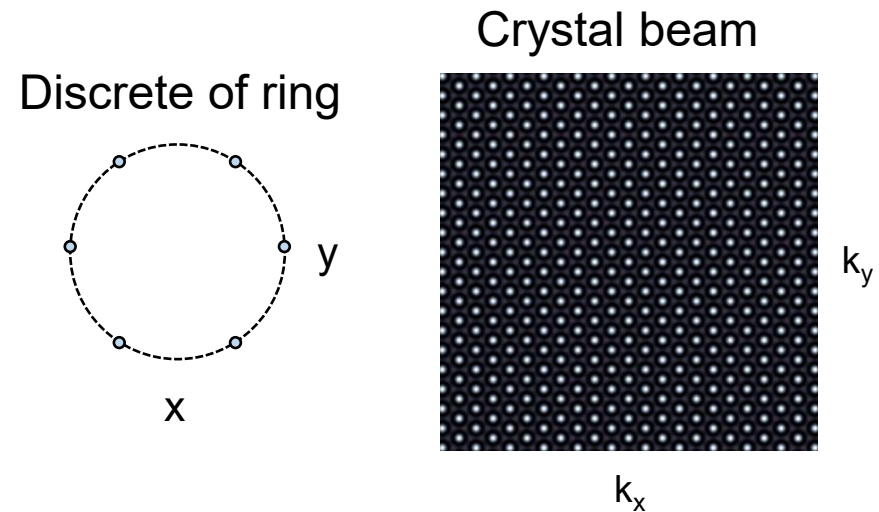
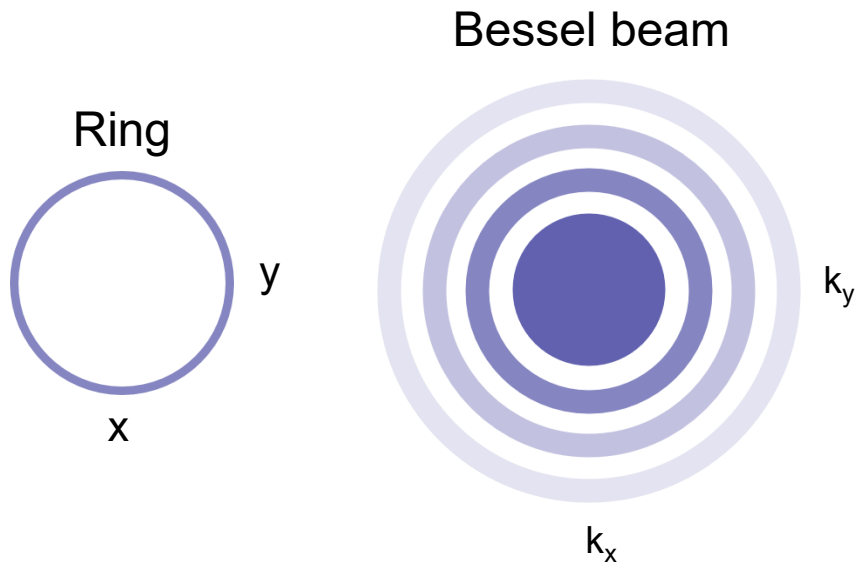
Contemp. Phys. 46, 15 (2005)

→ Fourier conjugate pair

Motivation

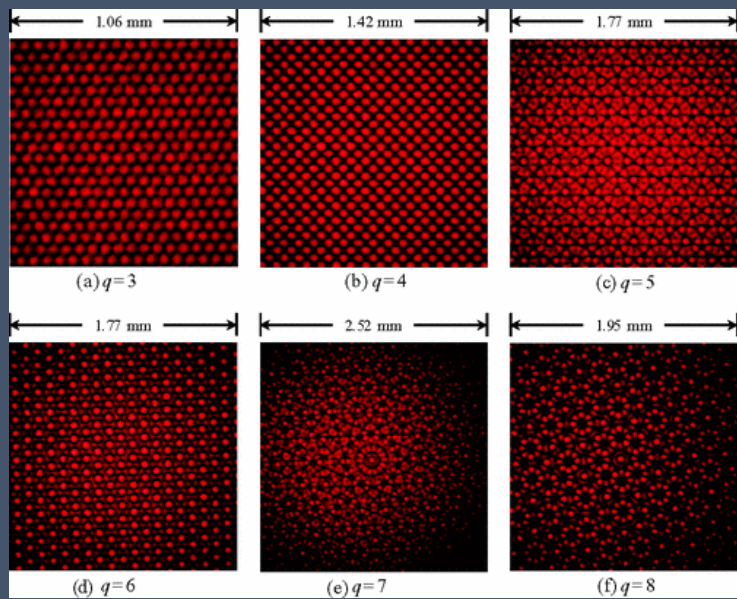
Motivation

Fourier conjugate pair

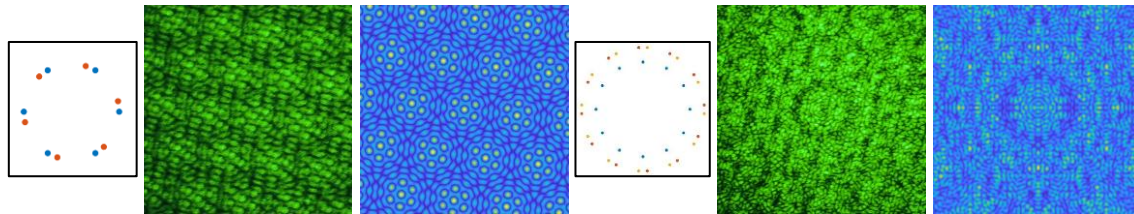
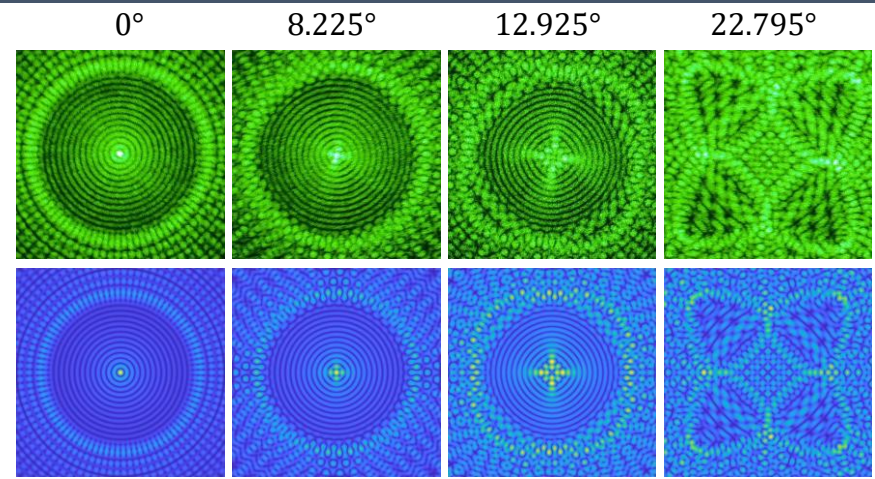
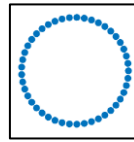


Motivation

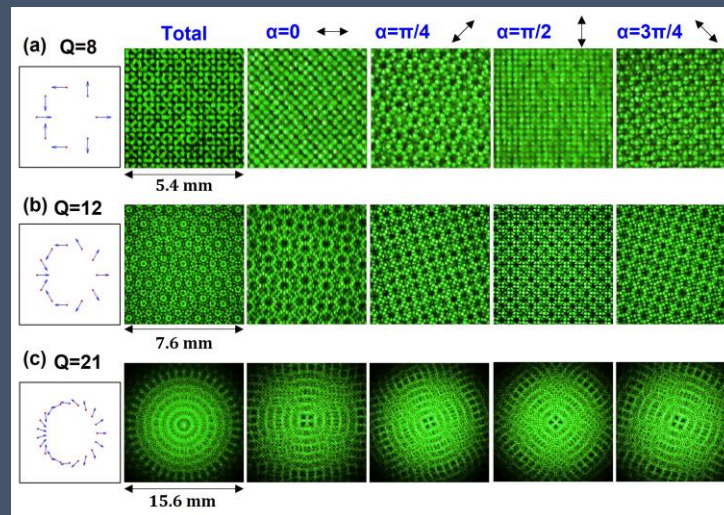
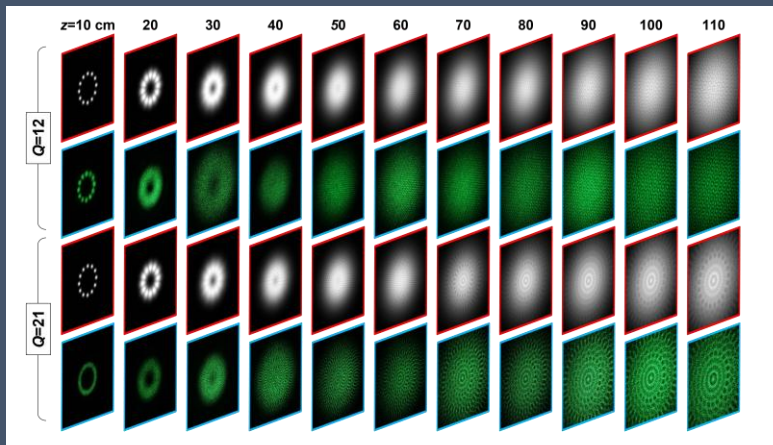
Phys. Rev. A 83, 053813



$q = 45$



by Y. F. Chen's group



by 游沅翰

by 黃凌琦



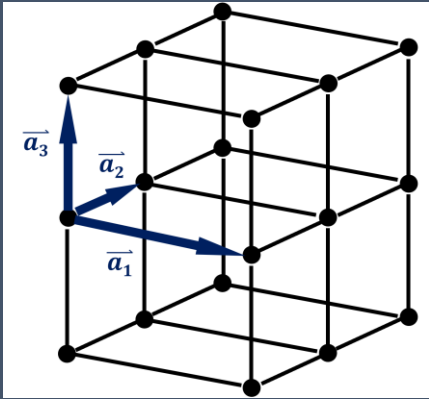
Optical and Acoustic Wave Manipulation Lab

Crystal & quasi-crystal structure



Crystal & Quasi-crystal structure

Crystal structure



Relation between basis:

$$\vec{R} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3, \in \mathbb{Z}$$

Translational symmetry:

$$F(\vec{r} + \vec{R}) = F(\vec{r})$$

Rotational symmetry (Q fold):

$$C_Q^S, \Delta\theta = \frac{2\pi}{Q} S$$

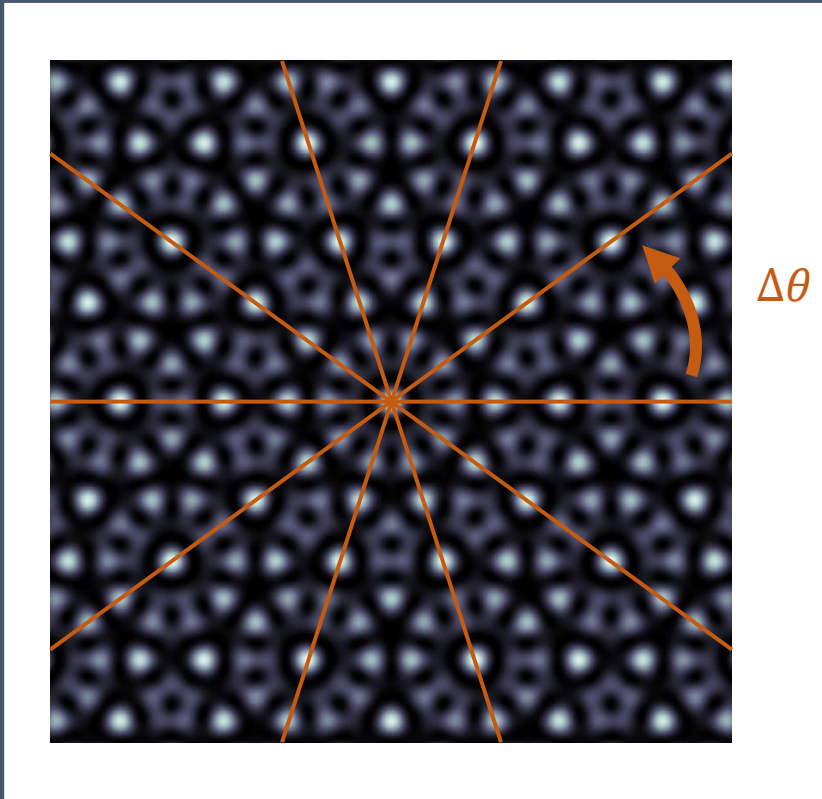


$Q = 2, 3, 4, 6$ fold



Crystal & Quasi-crystal structure

Quasi-crystal structure



Rotational symmetry:

$$C_{2Q}^S, \Delta\theta = \frac{2\pi}{2Q} S$$

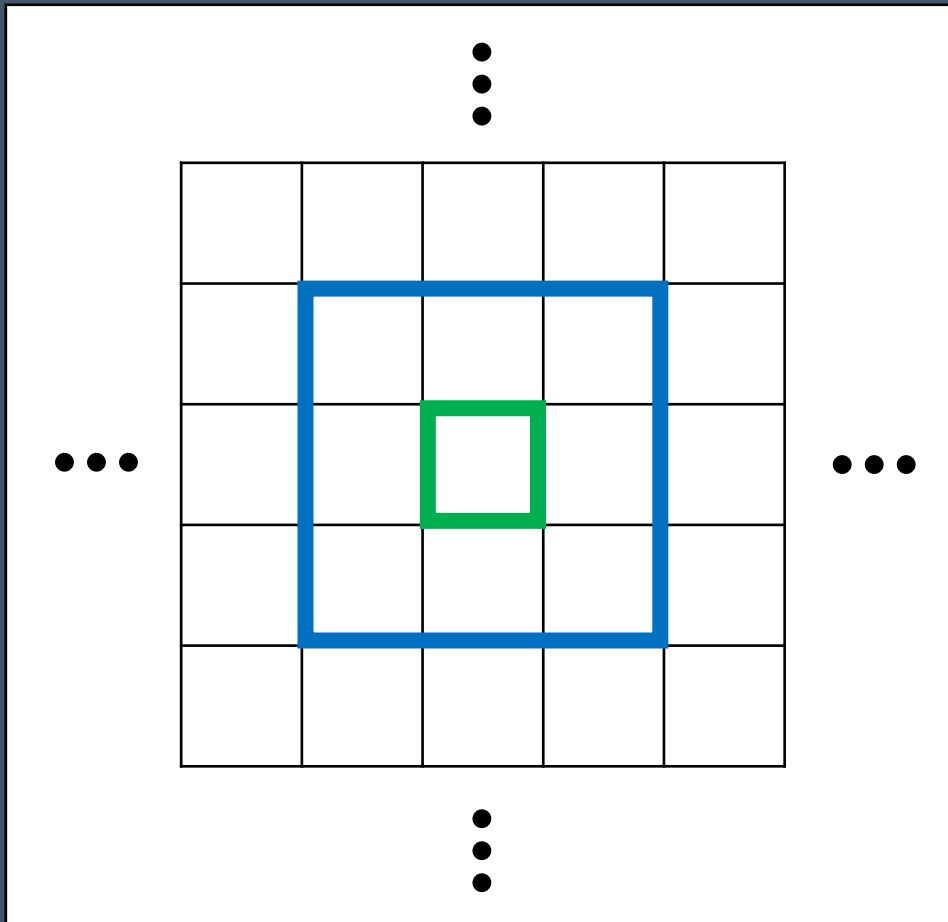
➡ $Q = 5, 7, 8, 9, \dots$ fold

Self-similar structure



Self-similar structure

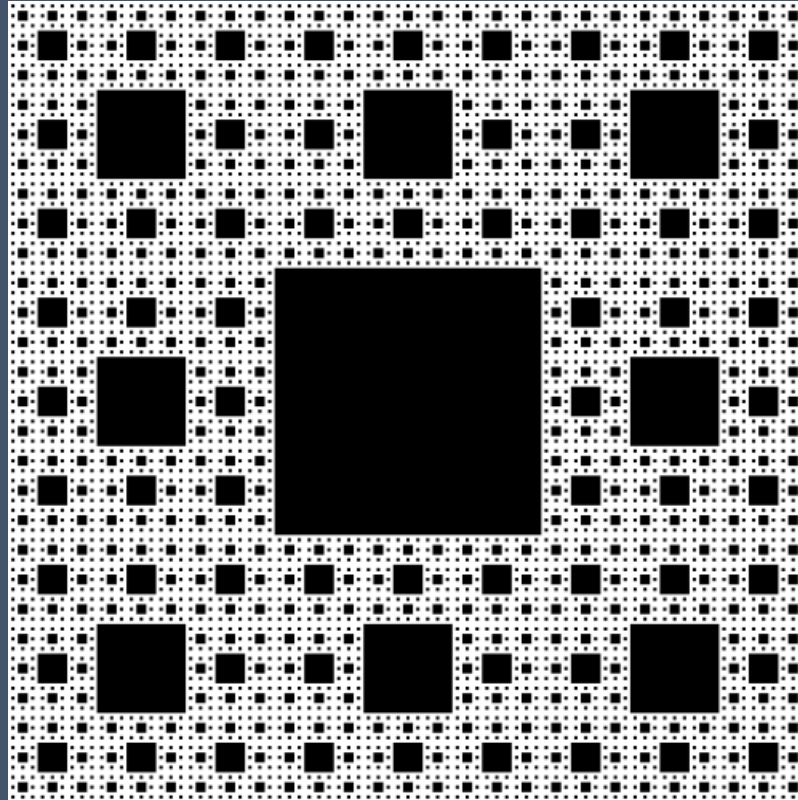
How many squares are there in the figure ?



**Similar structure
at any scale**

Self-similar structure

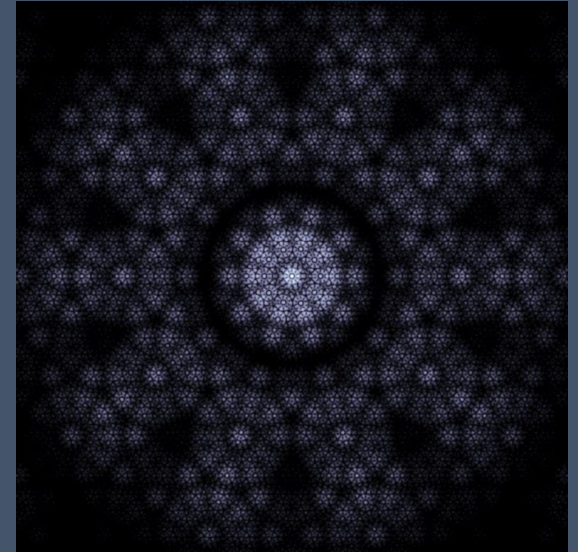
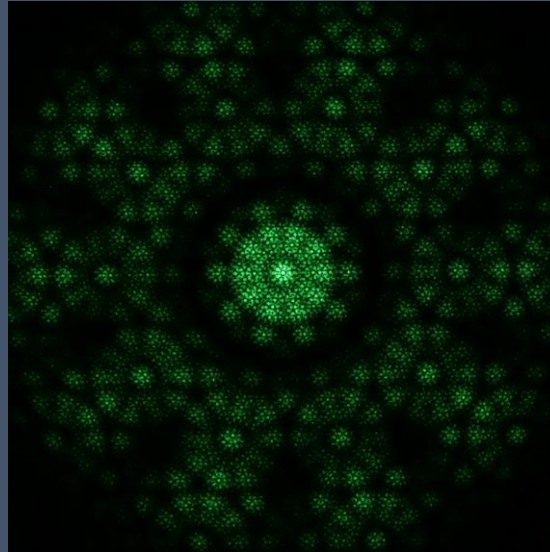
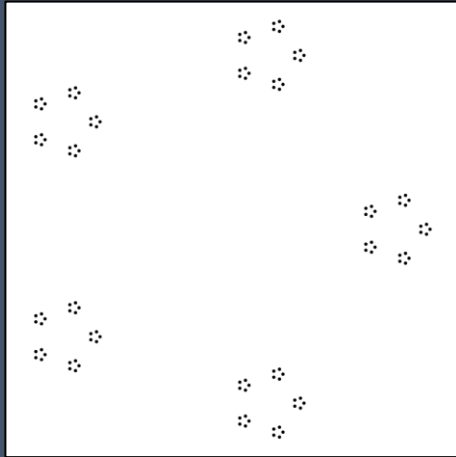
Sierpiński Carpet



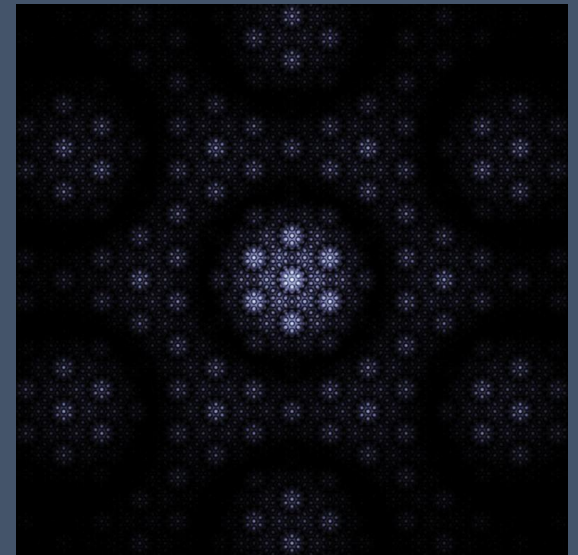
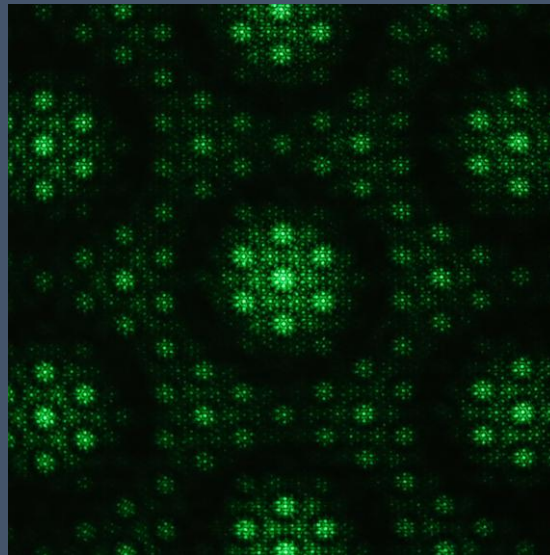
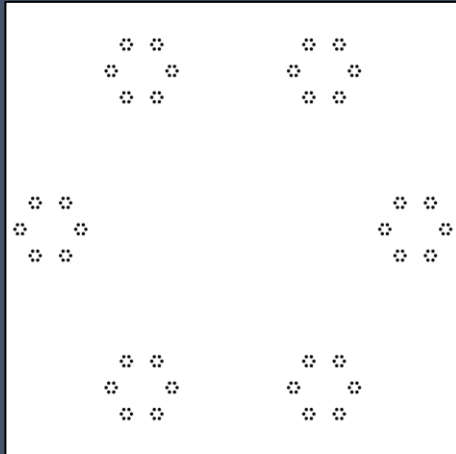
<https://zh.wikipedia.org/zh-tw/%E8%B0%A2%E5%B0%94%E5%AE%BE%E6%96%AF%E5%9F%BA%E5%9C%B0%E6%AF%AF>

Self-similar structure

Q=5



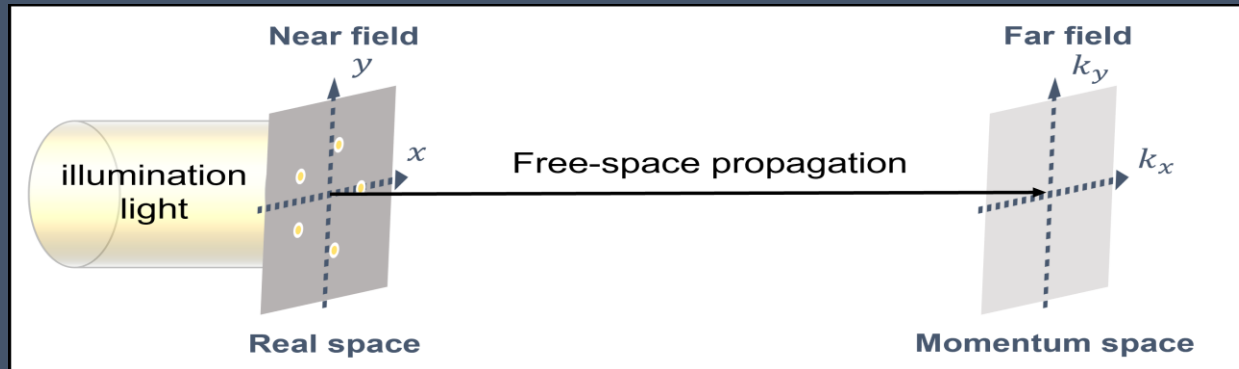
Q=6



Experimental principles



Fresnel diffraction & Fourier Optics



Fresnel diffraction integral (Near field):

$$\psi(x, y, z) = \frac{e^{ikz}}{i\lambda z} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(x_o, y_o, 0) e^{i\frac{k}{2z}[(x-x_o)^2 + (y-y_o)^2]} dx_o dy_o$$

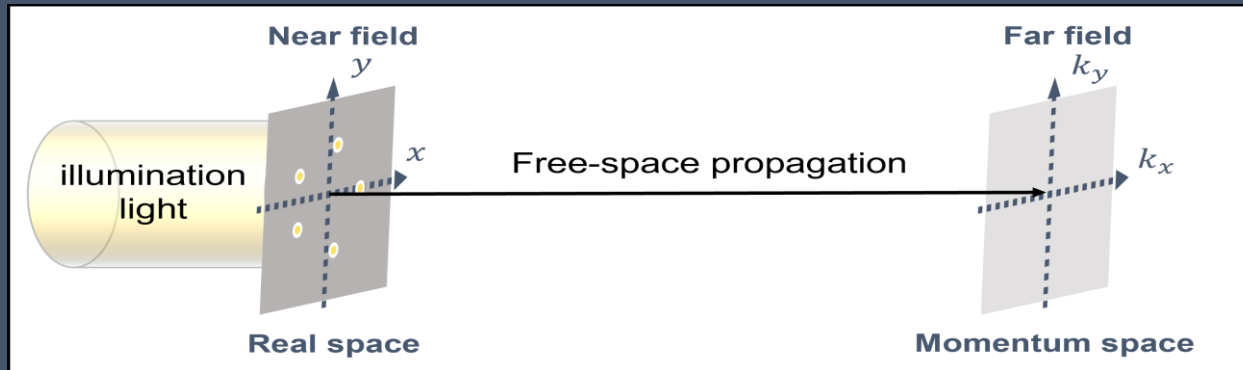
Far field approximation: $x_o \ll x, y_o \ll y$

Fraunhofer diffraction integral (Far field): $k_x = \frac{x}{z}k, k_y = \frac{y}{z}k$

$$\psi(x, y, z) = \frac{e^{ikz}}{i\lambda z} e^{i\frac{k}{2z}(x^2 + y^2)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(x_o, y_o, 0) e^{-ik_x x_o} e^{-ik_y y_o} dx_o dy_o$$



Fresnel diffraction & Fourier Optics



Fraunhofer diffraction integral:

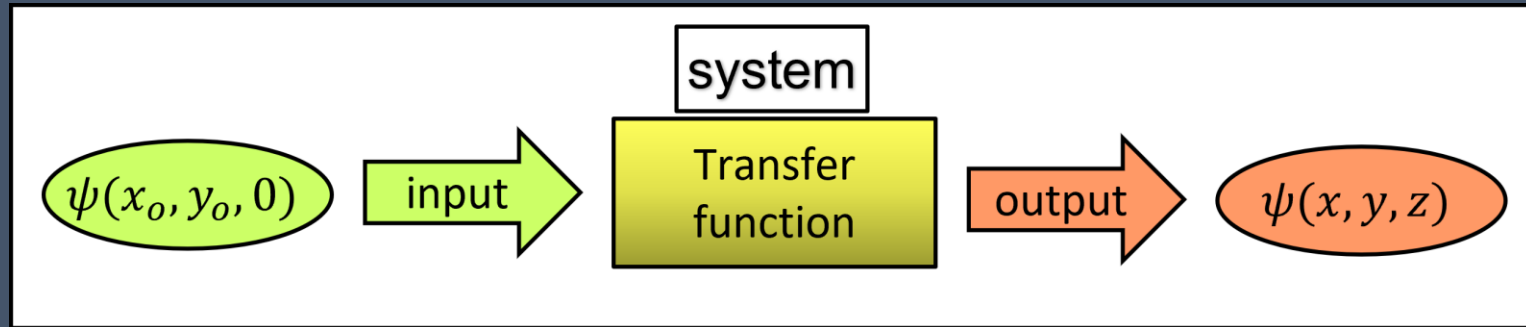
$$\psi(x, y, z) = \frac{e^{ikz}}{i\lambda z} e^{i\frac{k}{2z}(x^2+y^2)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(x_o, y_o, 0) e^{-ik_x x_o} e^{-ik_y y_o} dx_o dy_o$$

$$\rightarrow \psi(x, y, z) = \frac{e^{ikz}}{i\lambda z} e^{i\frac{k}{2z}(x^2+y^2)} \boxed{F.T.^{-1} \{ \psi(x_o, y_o, 0) \}}$$

↓

$$\tilde{\psi}(k_x, k_y)$$

Fresnel diffraction & Fourier Optics



Transfer function:

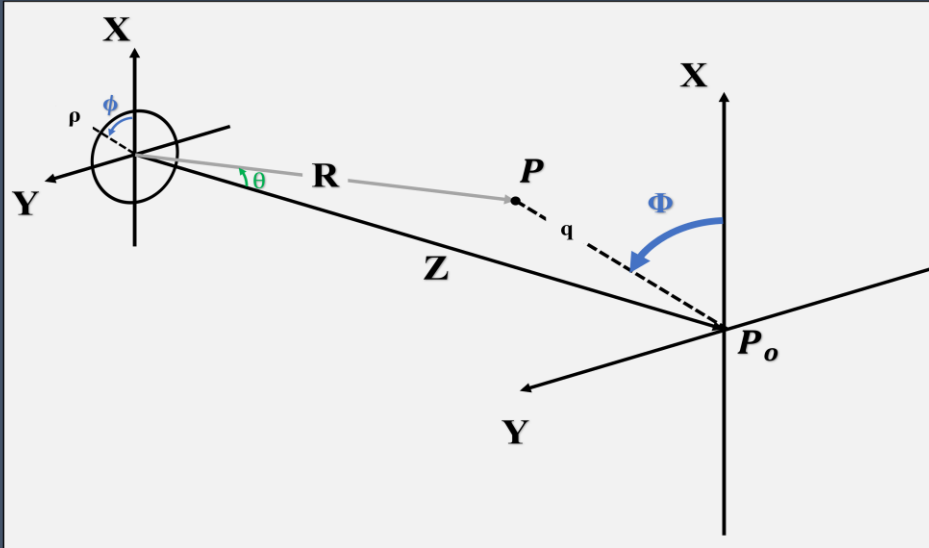
$$h(x - x_o, y - y_o, z) = \frac{e^{ikz}}{i\lambda z} e^{i\frac{k}{2z}[(x-x_o)^2 + (y-y_o)^2]}$$

Fresnel diffraction integral:

$$\psi(x, y, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \psi(x_o, y_o, 0) h(x - x_o, y - y_o, z) dx_o dy_o$$

$$\rightarrow \psi(x, y, z) = (\psi_o * h)(x, y, z)$$

Aperture diffraction



Huygens principle:

$$\vec{E} = \frac{\varepsilon_A}{R} e^{i(\omega t - kR)} \iint e^{ik(Xx + Yy)} dS$$

ε_A : intensity of incident light

ω : frequency of incident light

k : wave number

$$x = \rho \cos \phi \quad y = \rho \sin \phi \quad dS = \rho d\rho d\phi$$

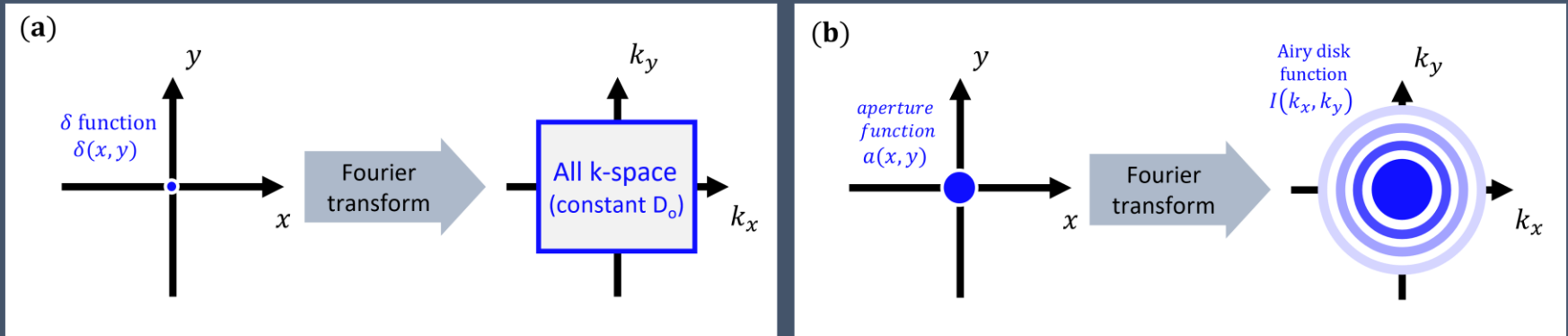
$$X = q \cos \Phi \quad Y = q \sin \Phi$$

Rotational symmetry:
$$\vec{E} = \frac{\varepsilon_A}{R} e^{i(\omega t - kR)} \int_{\rho=0}^a \int_{\phi=0}^{2\pi} \rho e^{\frac{ikq}{R} \rho \cos \phi} d\rho d\phi$$

Bessel function integral form:
$$\vec{E} = \frac{2\pi\varepsilon_A}{R} e^{i(\omega t - kR)} \int_{\rho=0}^a J_0\left(\frac{k\rho q}{R}\right) \rho d\rho$$

Airy disk function:
$$I(\theta) = I_0 \left[\frac{2J_0(ka \sin \theta)}{ka\theta} \right]^2$$

Aperture diffraction



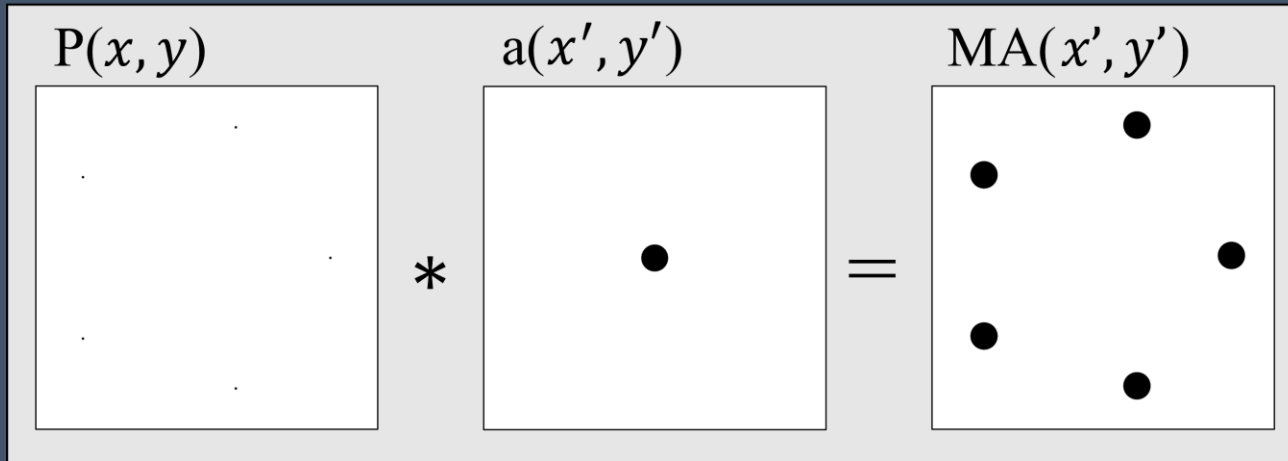
Fourier pair of δ function: $F.T \{ \delta(x, y) \} = D_o$

Fourier pair of aperture function: $F.T \{ a(x, y) \} = I(k_x, k_y)$

Convolution theorem: $\delta * a(x, y) = F.T.^{-1} \{ D_o \cdot I(k_x, k_y) \}$

➡ Putting aperture on the specific position.

Aperture diffraction



Position function: $P(x, y) = \sum_j \delta_j(x_j, y_j)$

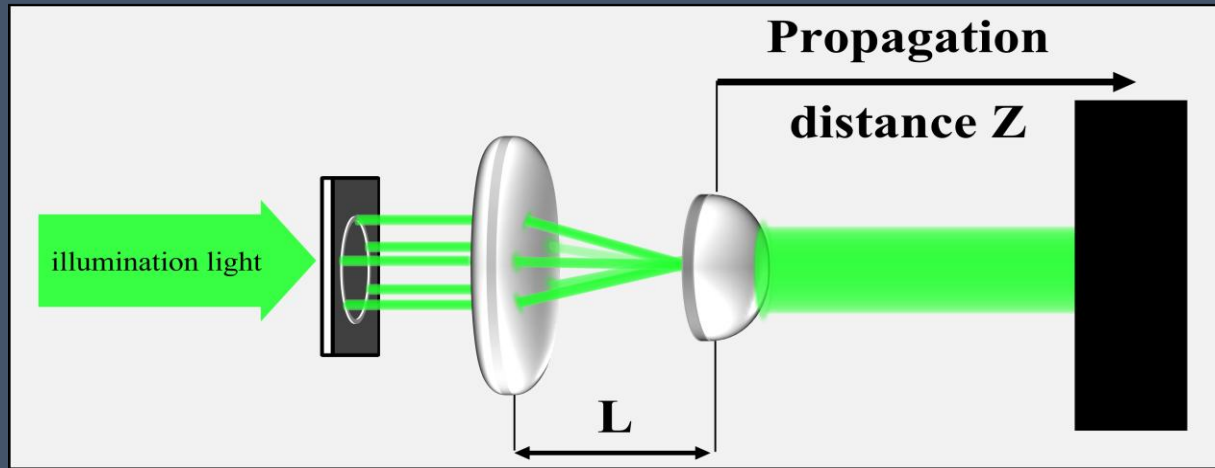
Single aperture function: $a(x', y')$

Multiple aperture function:

$$MA(x', y') = P * a(x', y') = \sum_j a_j(x'_j, y'_j)$$



Multi-beam interference



Gaussian beam wave form:

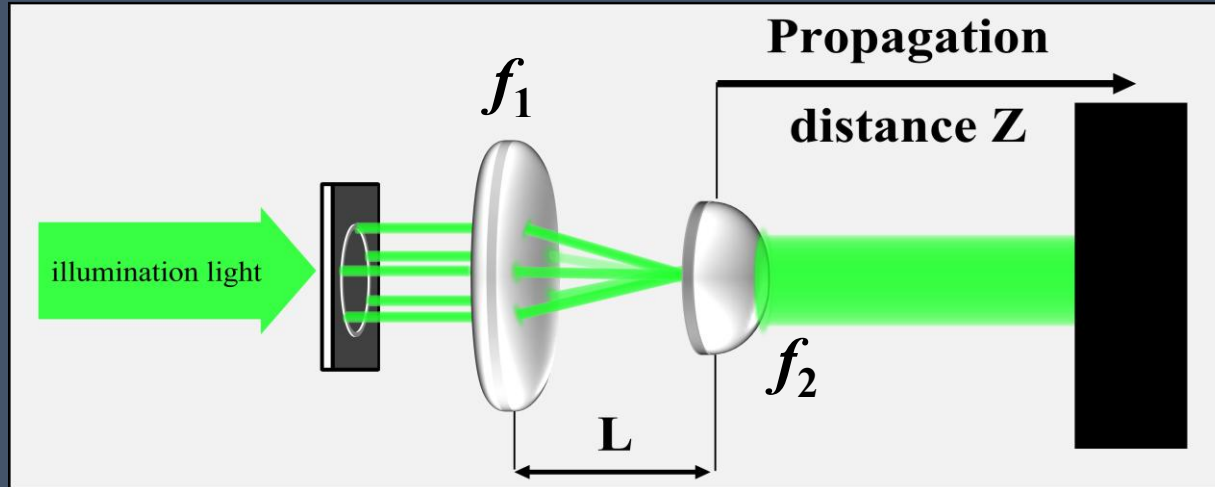
$$\psi(x_o, y_o, 0) = \sqrt{\frac{1}{Q}} \sqrt{\frac{2}{\pi \omega_o^2}} \sum_{j=0}^{Q-1} \exp \left[-\frac{(x_o - x_j)^2 + (y_o - y_j)^2}{\omega_o^2} \right]$$

position of aperture: $(x_j, y_j) = (\rho_j \cos \theta_j, \rho_j \sin \theta_j)$ $\theta_j = \frac{2\pi j}{Q}$

ω_o : beam waist

Q : number of Gaussian beam

Multi-beam interference



ABCD matrix:
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1 & z \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_2} & 1 \end{bmatrix} \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{1}{f_1} & 1 \end{bmatrix}$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \left(1 - \frac{L}{f_1}\right)\left(1 - \frac{L}{f_2}\right) - \frac{z}{f_1} & L + z - \frac{zL}{f_2} \\ \frac{L}{f_1 f_2} - \frac{1}{f_1} - \frac{1}{f_2} & 1 - \frac{L}{f_2} \end{bmatrix}$$



Multi-beam interference

wavefront curvature:

$$R(z) = A \left[(A \cdot z_R)^2 + B^2 \right] \quad z_R : \text{Rayleigh length}$$

beam radius:

$$\omega(z) = \omega_o \sqrt{A^2 + (B/z_R)^2}$$

complex beam parameter: $q = z - iz_R$

$$q(z_2) = \frac{A \cdot q(z_1) + B}{C \cdot q(z_1) + D} \quad \frac{1}{q(z)} = \frac{1}{R(z)} + i \frac{\lambda}{\pi \omega^2(z)}$$

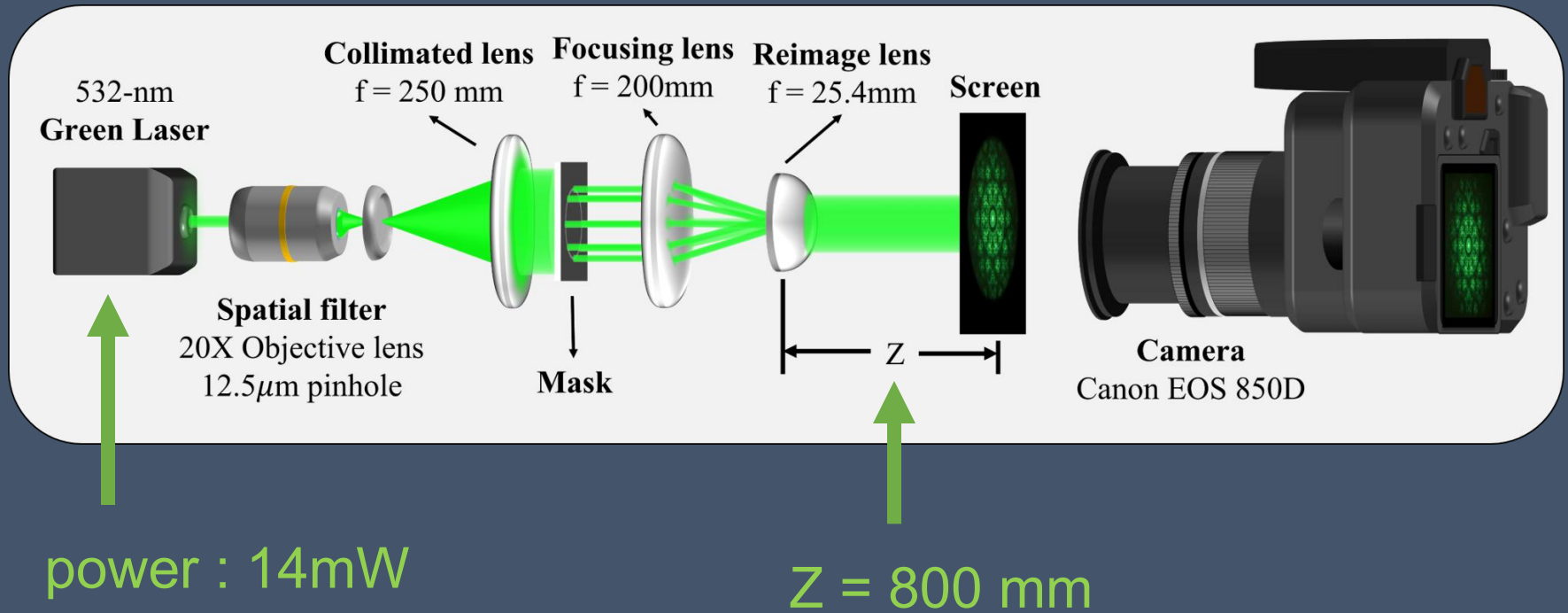
Fresnel-Huygens theory:

$$\psi(x, y, z) = e^{i\xi(x, y, z)} \sum_{j=1}^Q \exp \left\{ -ik \left[\frac{(x - Ax_j)^2 + (y - Ay_j)^2}{2q(z)} \right] \right\}$$

$$\xi(x, y, z) = k \left\{ z + \left[C(x^2 + y^2)/(2A) \right] \right\}$$

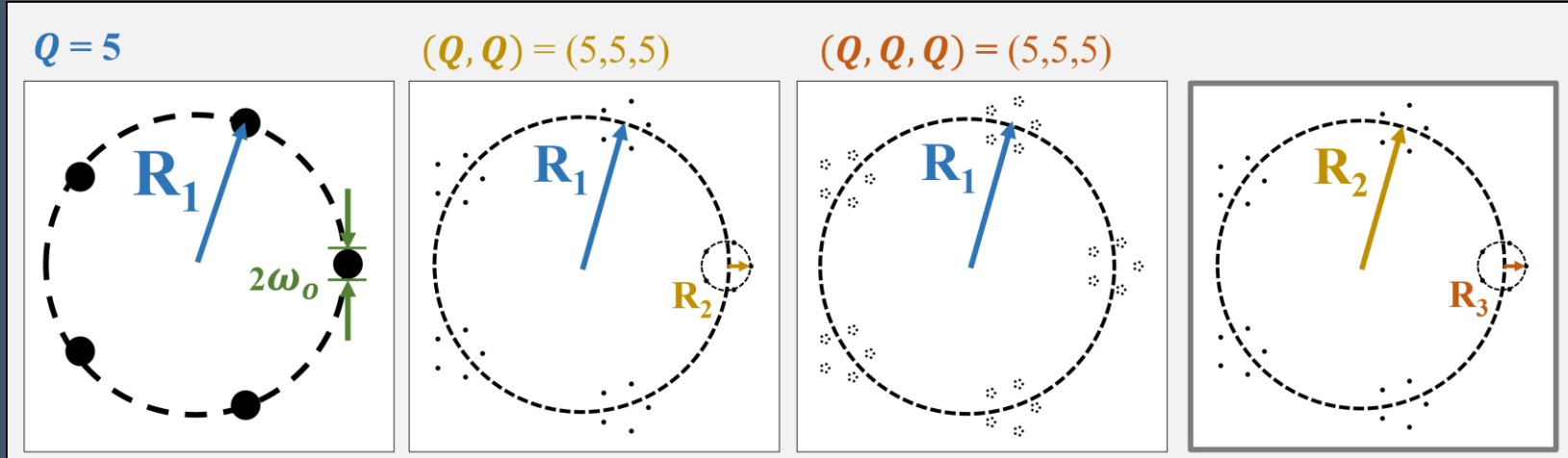
Experimental setup

Experimental Setup



Photomask

Photomask: configurations & fabrication results



Aperture position in the **single-layer** mask : $(x_\alpha, y_\alpha) = (R_1 \cos \theta_\alpha, R_1 \sin \theta_\alpha)$

Aperture position in the **double-layer** mask :

$$(x_{\alpha\beta}, y_{\alpha\beta}) = (R_1 \cos \theta_\alpha + R_2 \cos \theta_\beta, R_1 \sin \theta_\alpha + R_2 \sin \theta_\beta)$$

Aperture position in the **triple-layer** mask :

$$(x_{\alpha\beta\gamma}, y_{\alpha\beta\gamma}) = (R_1 \cos \theta_\alpha + R_2 \cos \theta_\beta + R_3 \cos \theta_\gamma, R_1 \sin \theta_\alpha + R_2 \sin \theta_\beta + R_3 \sin \theta_\gamma)$$

$$\theta_\alpha = \frac{2\pi}{Q} \alpha, \quad \alpha = 0, 1, 2, \dots, Q-1; \quad \theta_\beta = \frac{2\pi}{Q} \beta, \quad \beta = 0, 1, 2, \dots, Q-1; \quad \theta_\gamma = \frac{2\pi}{Q} \gamma, \quad \gamma = 0, 1, 2, \dots, Q-1;$$

$$\omega_0 = 0.1, \quad R_1 = 6, \quad R_2 = 1, \quad R_3 = \frac{1}{6} \text{ (mm)}.$$



Photomask: configurations & Fabrication results

1st Gen.
outsourcing



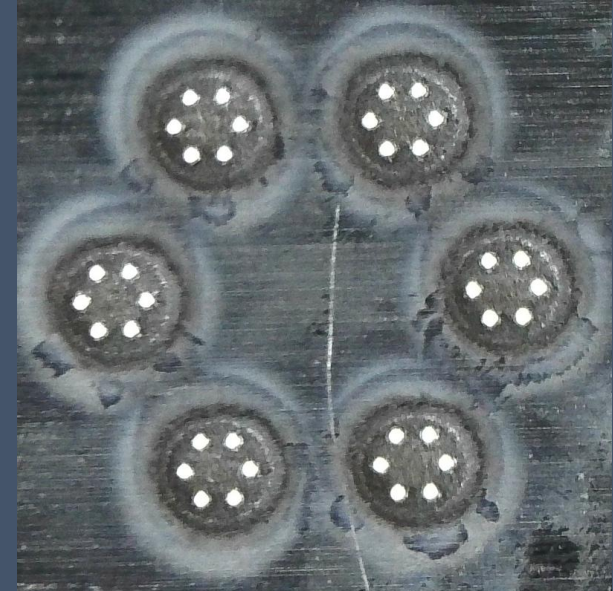
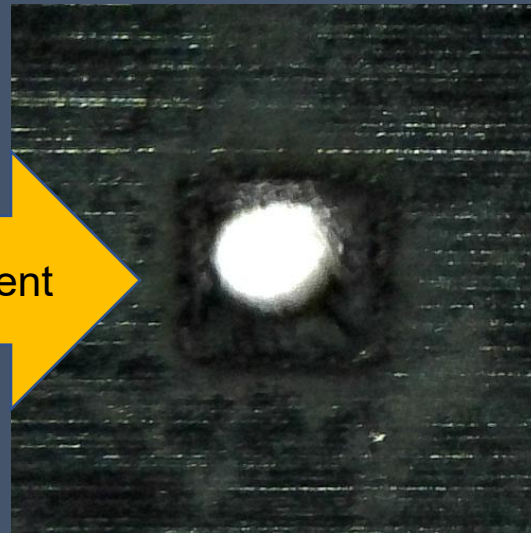
2nd Gen.



Aluminum sheet:



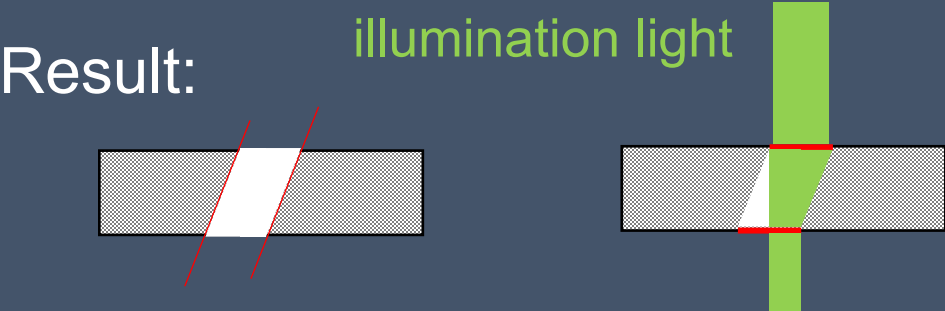
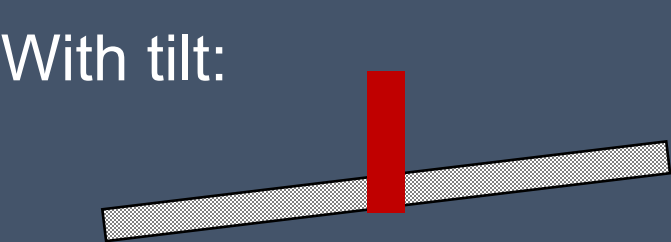
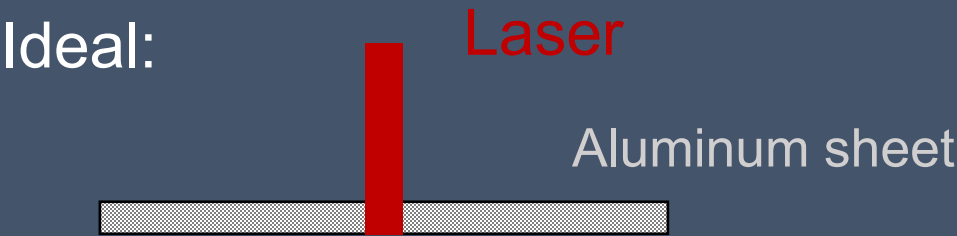
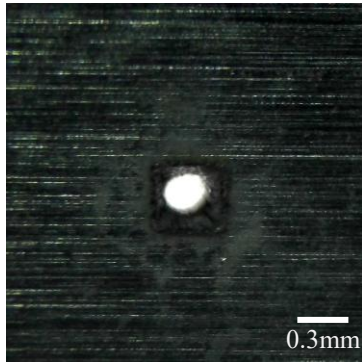
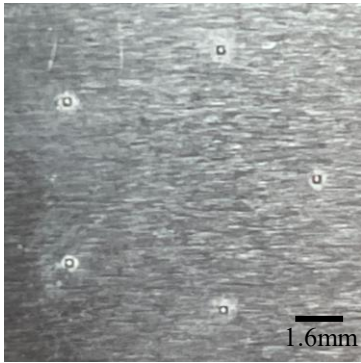
Improvement



Photomask: configurations & Fabrication results



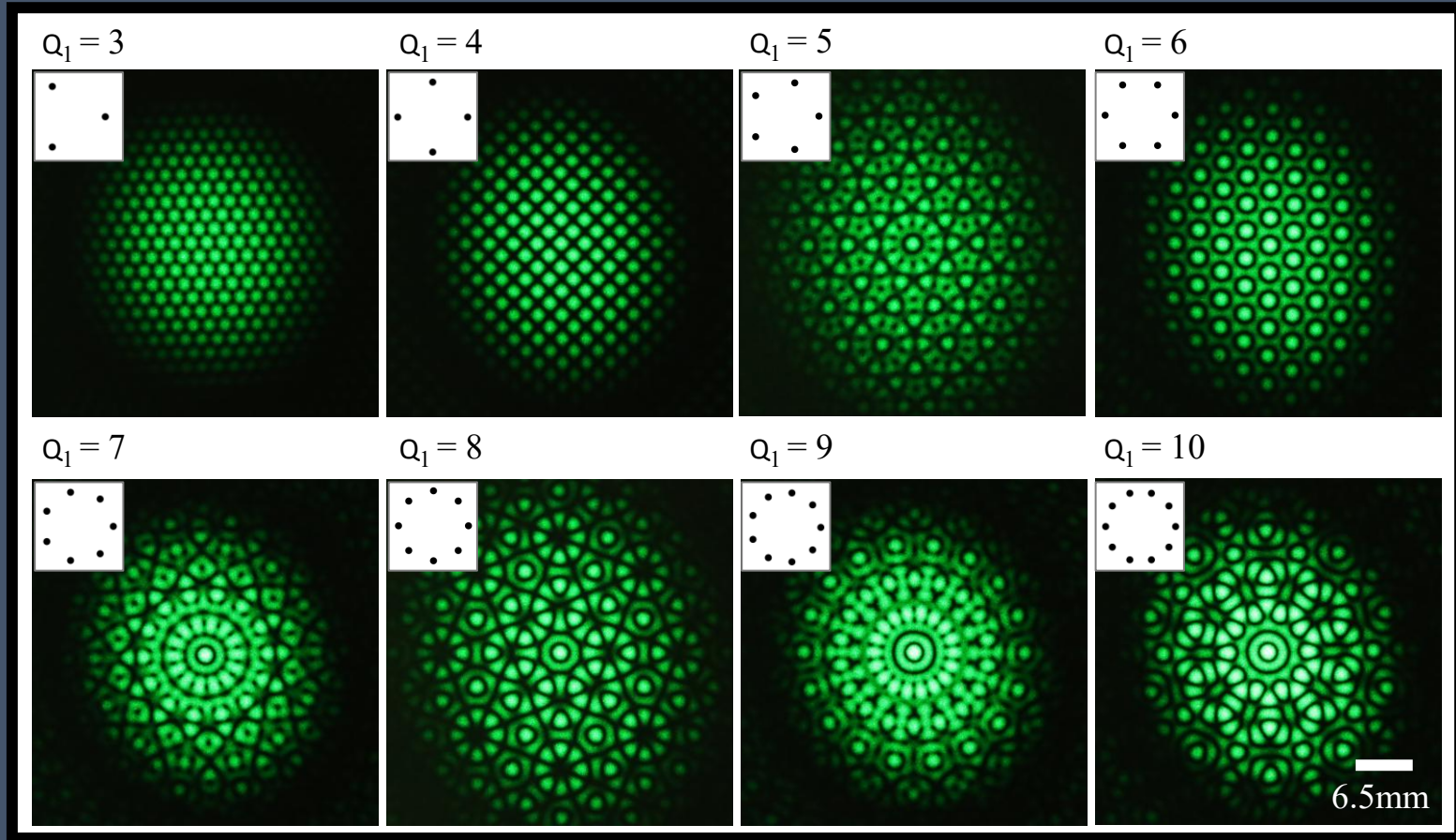
- Wavelength: 1066 nm (for metal)
- Power: 30 W @ 20 kHz
- X-Y resolution: 10 μ m
- Beam width: 30 μ m
- Working distance: 336 mm
- Dof: 30 μ m
- Image format: dxf or internal SW



Experimental results & Discussion

Experimental results

Different number of source:

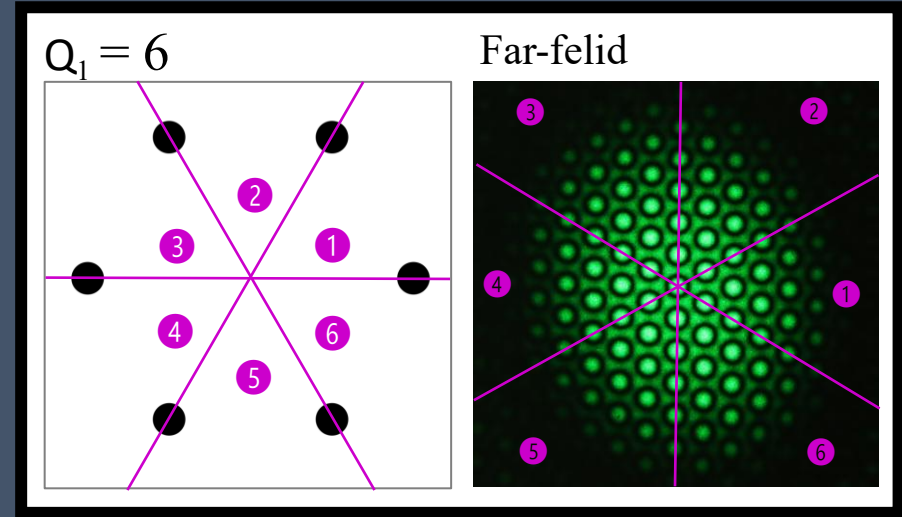
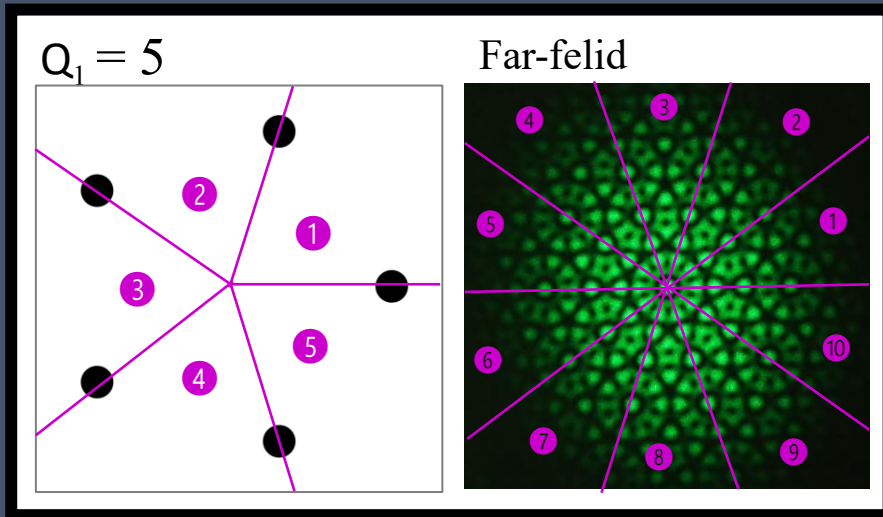


Identity structure for each Q_1



Experimental results

Rotational symmetry



Far Field Symmetry Axis Number:

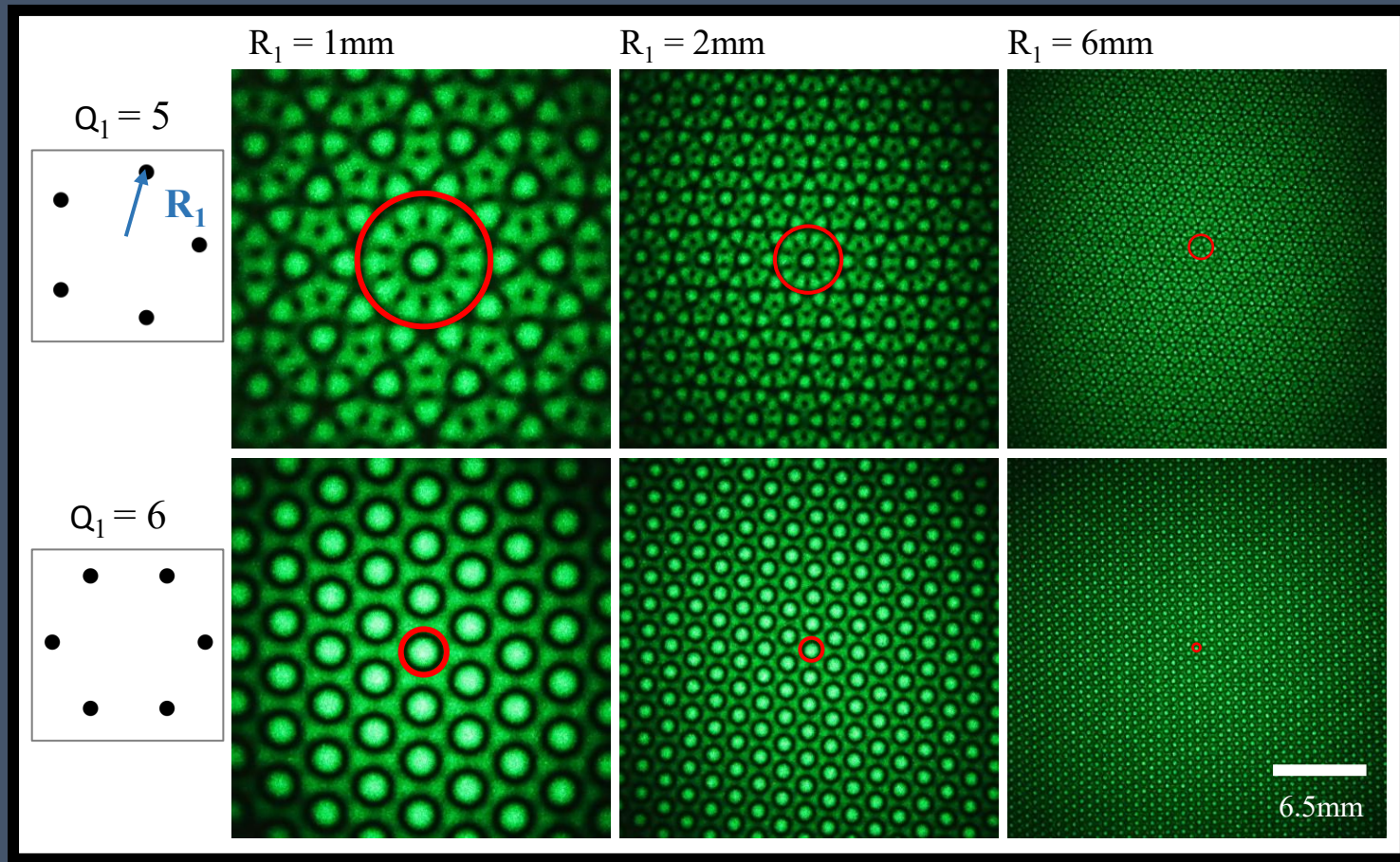
$$Q_1 = \text{odd}, \text{FFSAN} = 2Q_1$$

$$Q_1 = \text{even}, \text{FFSAN} = Q_1$$



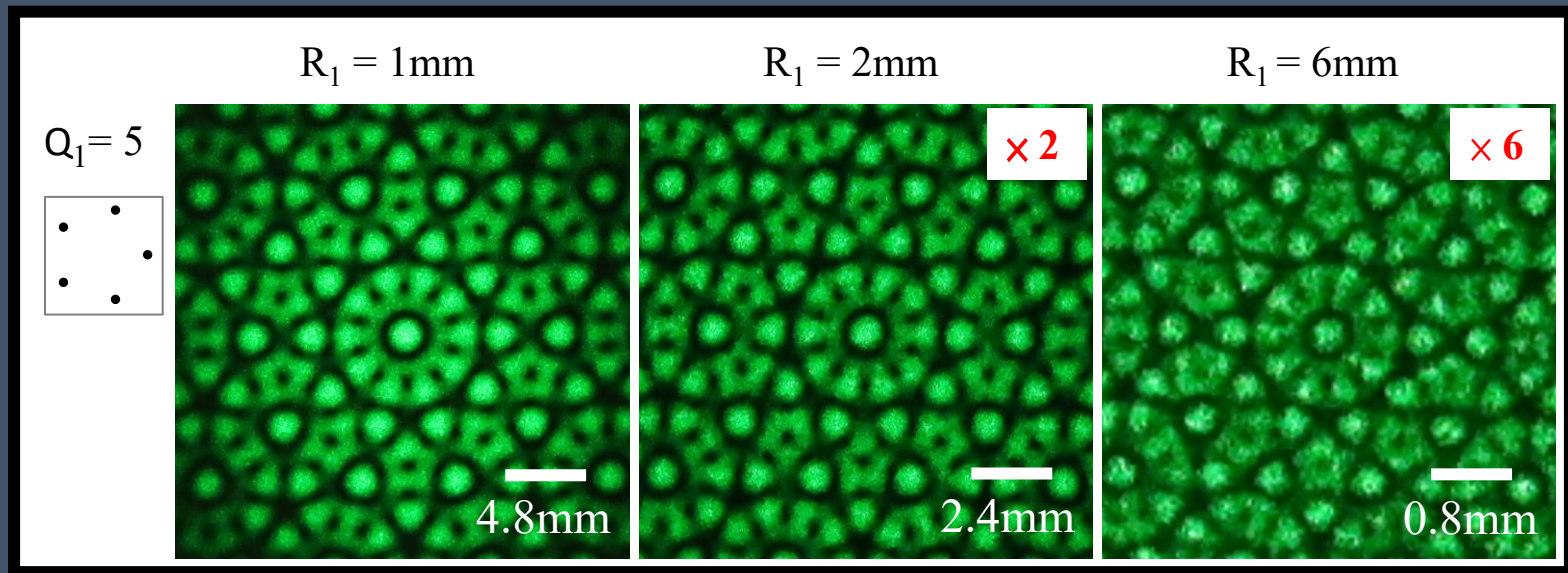
Experimental results

Different distance between sources:



Experimental results

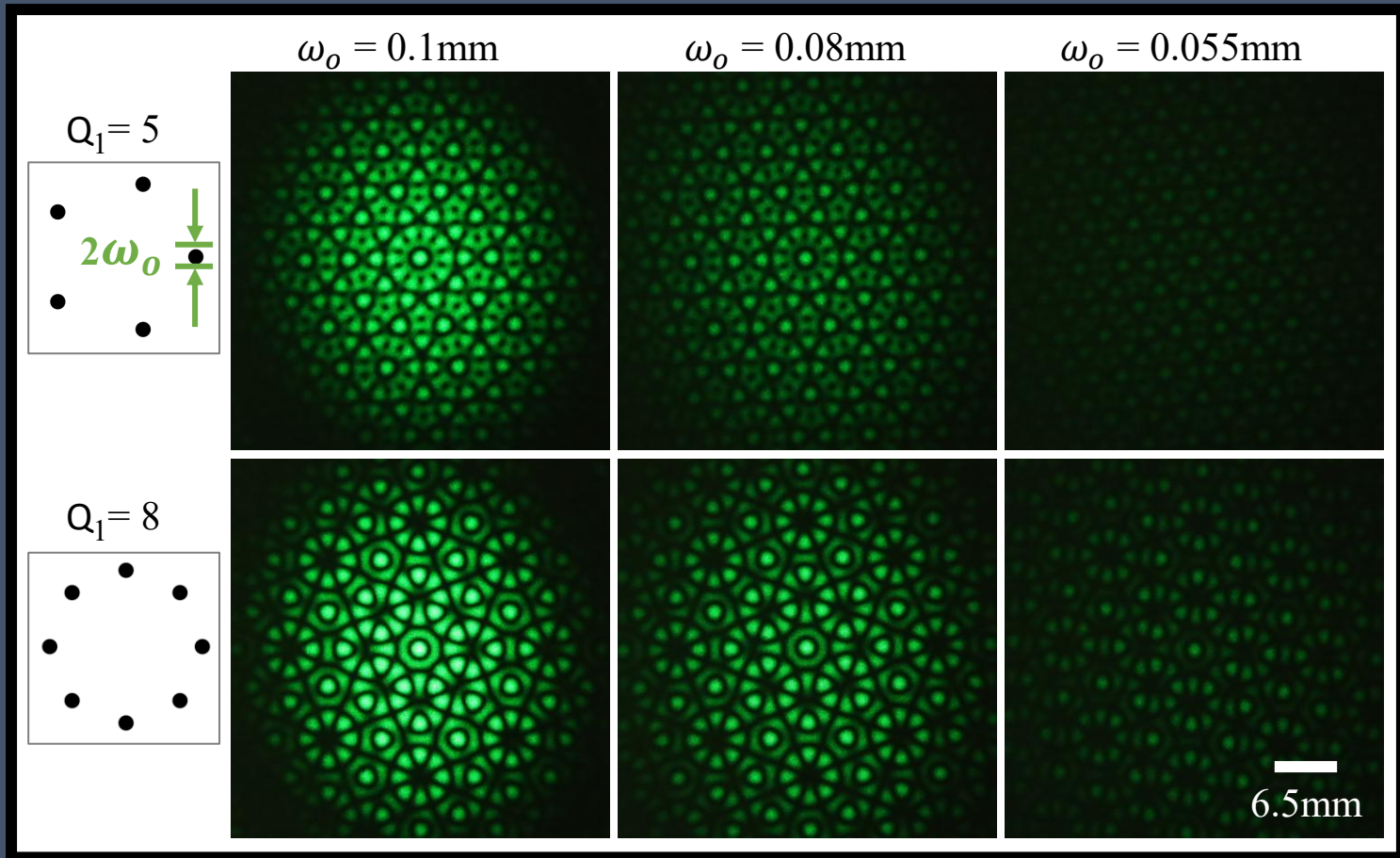
Different distance between sources:



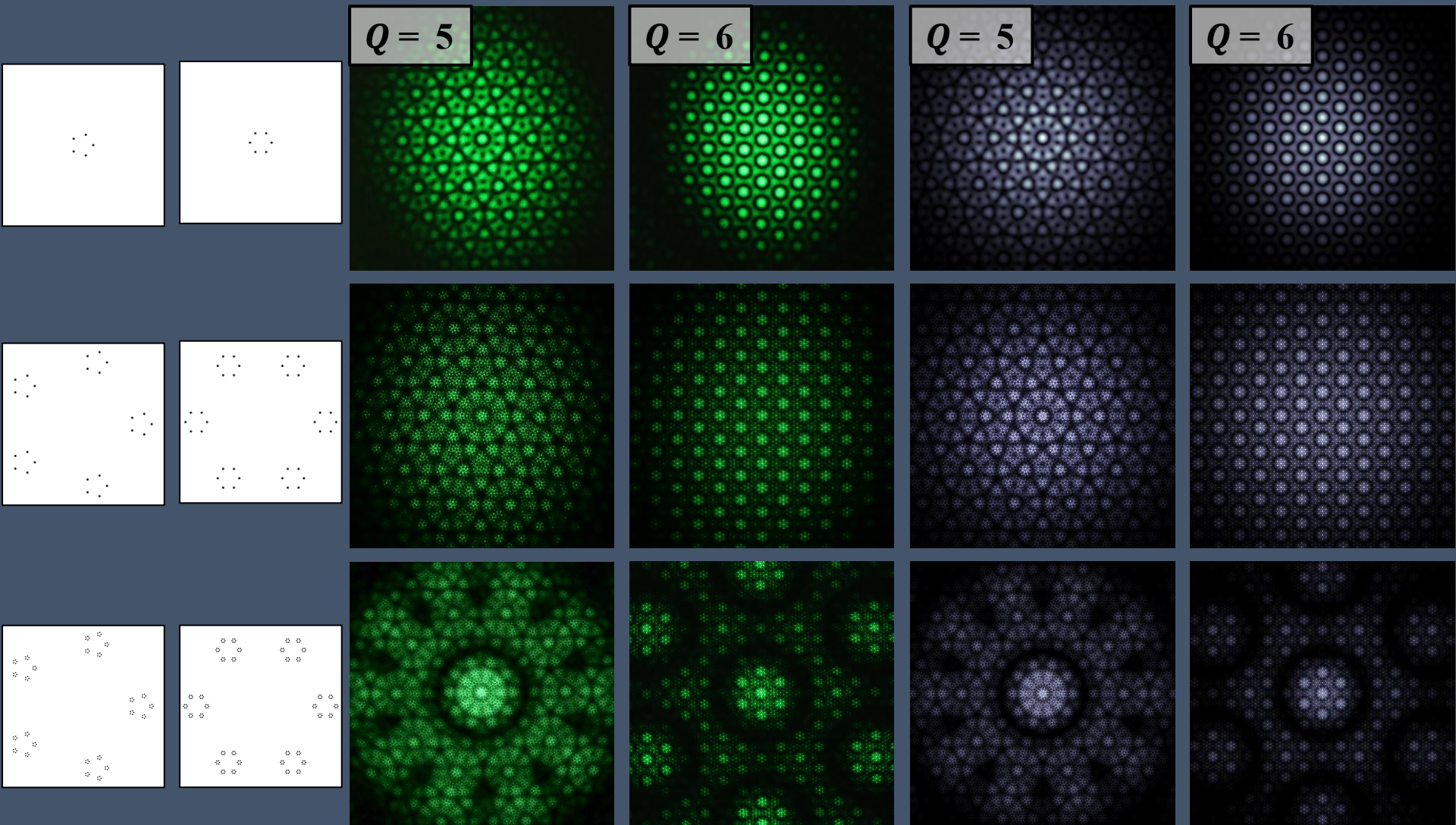
Identity structure for $Q_1 = 5$

Experimental results

Different beam radius of sources:

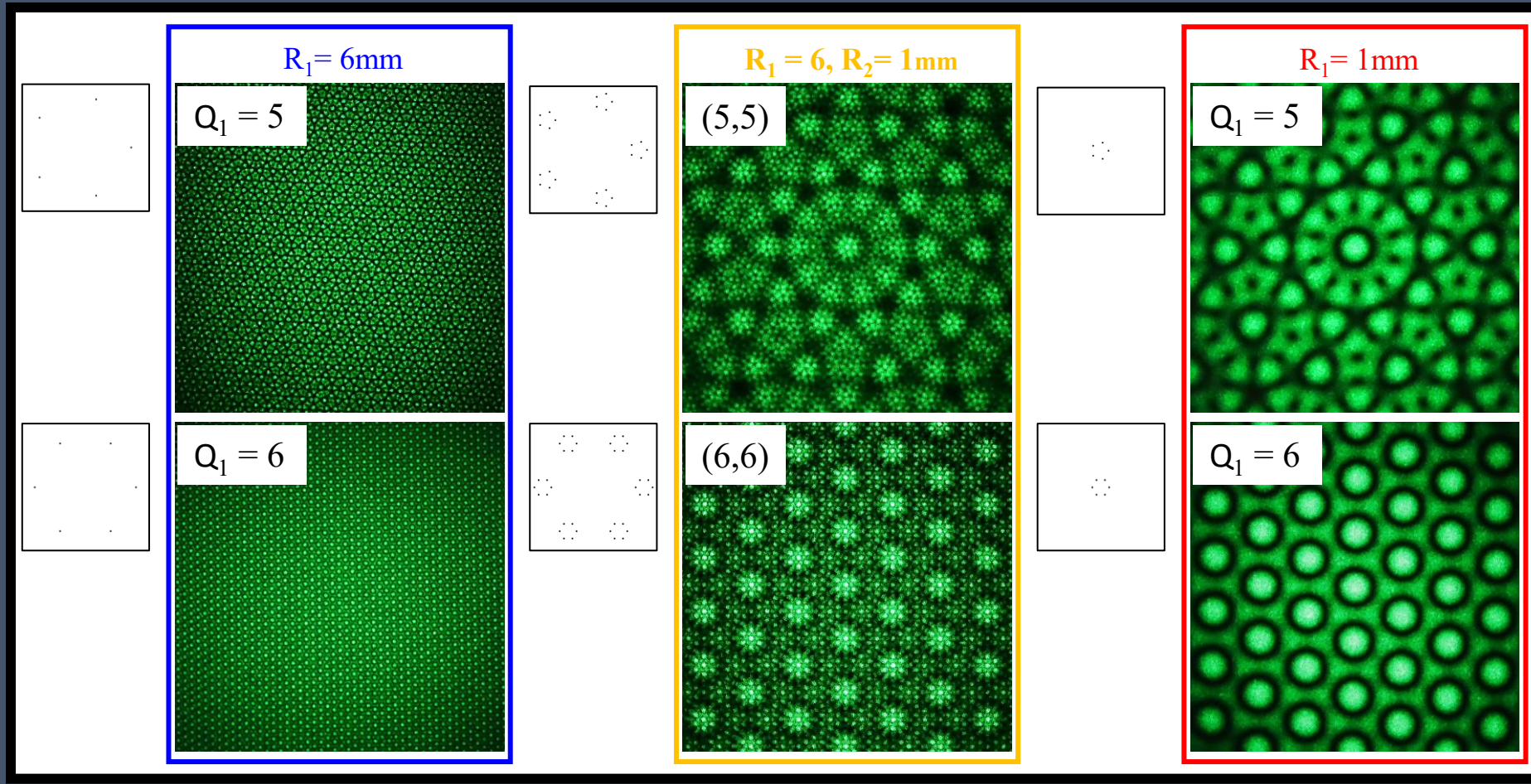


Reconstruction



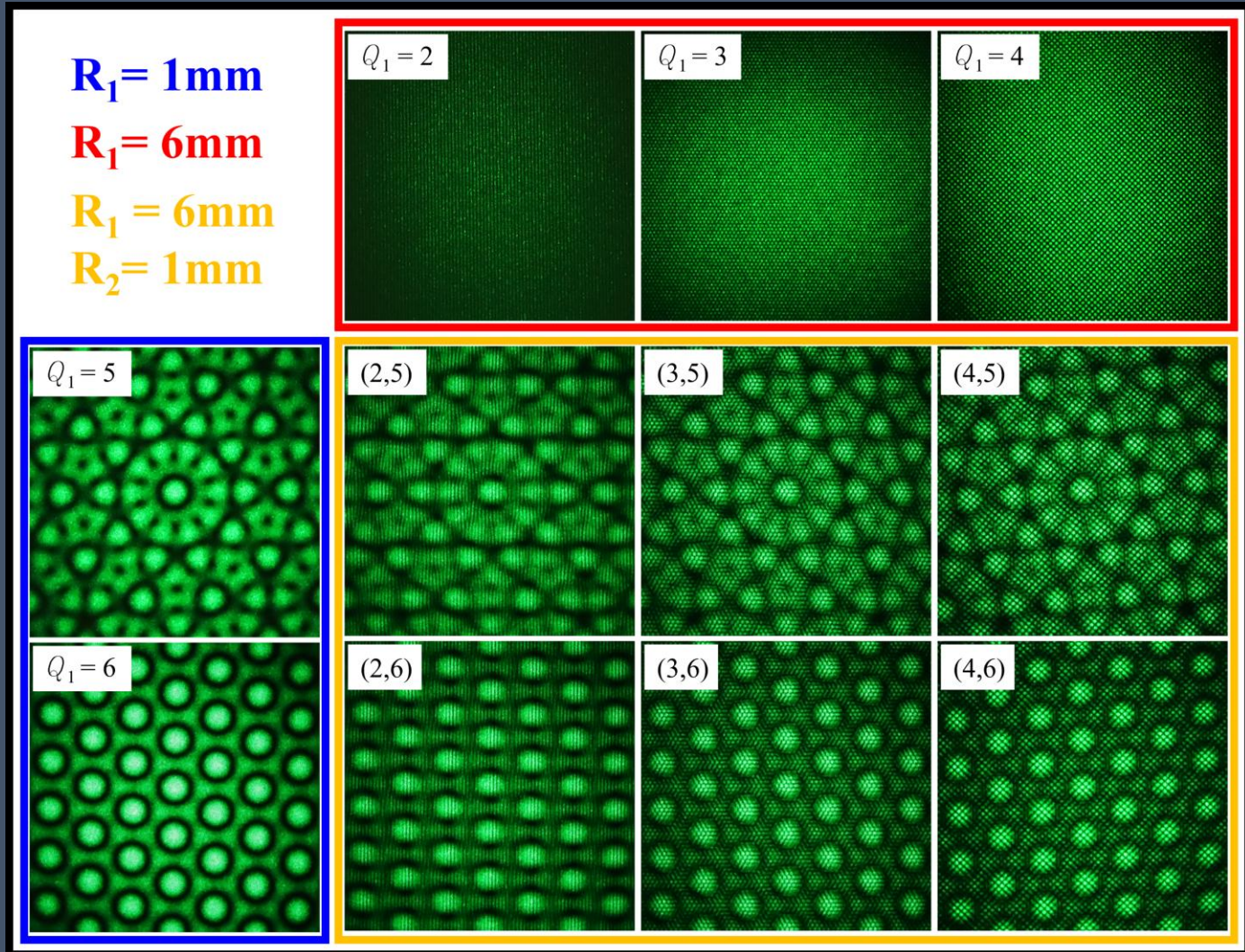
Discussion

Comparison between single layer & double layer:



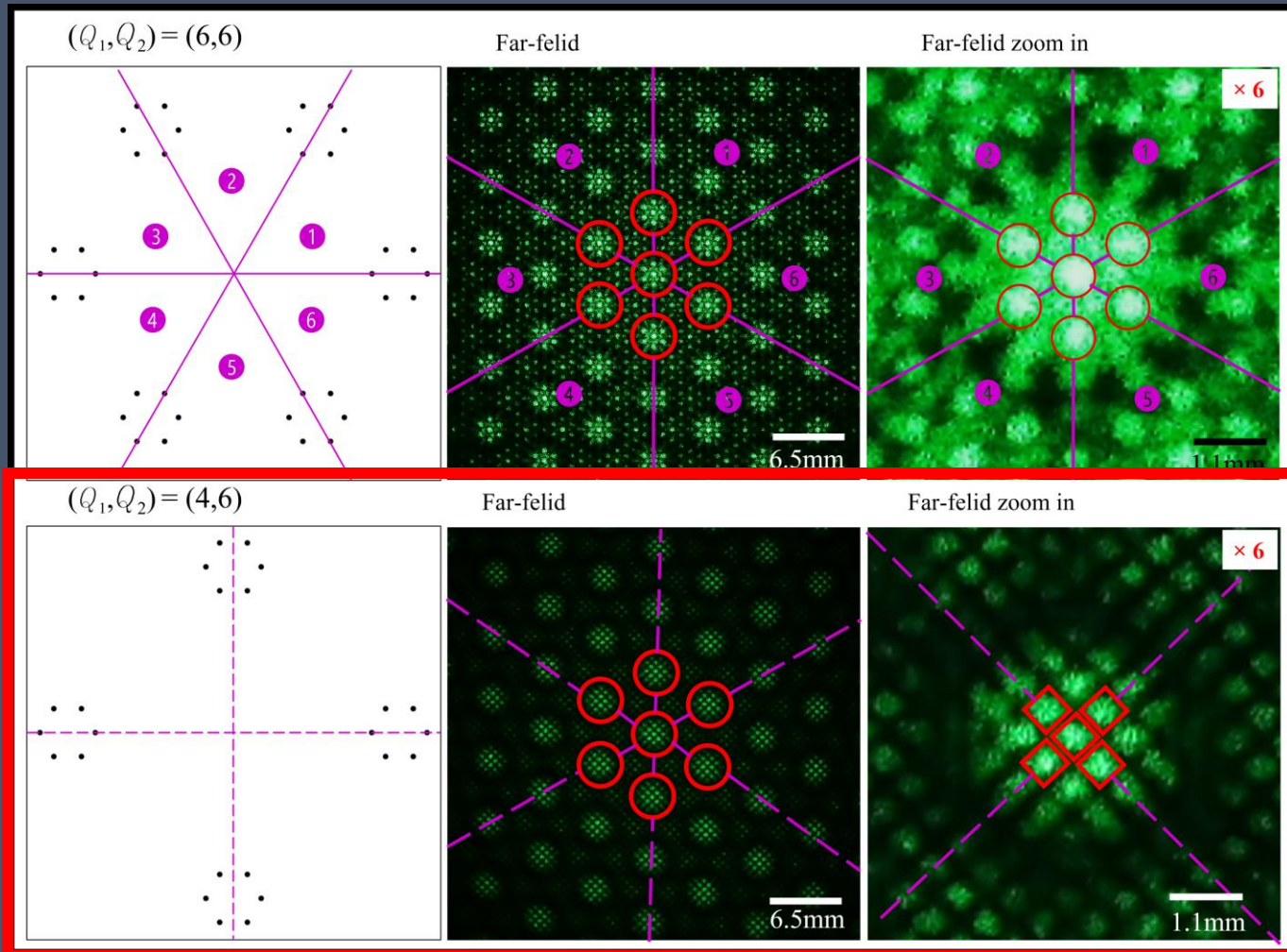
Discussion

Comparison between single layer & double layer:



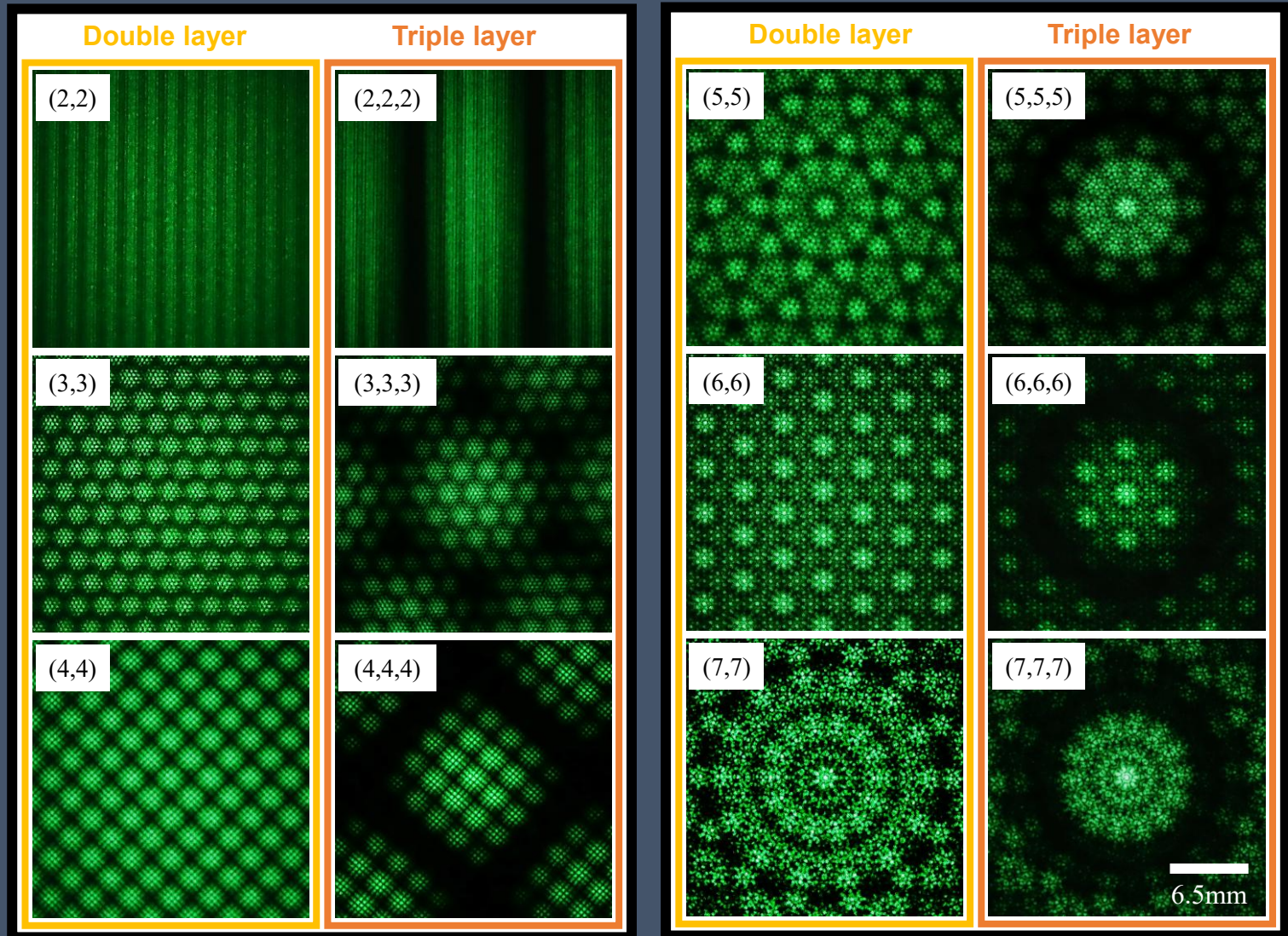
Discussion

Rotational symmetry



Discussion

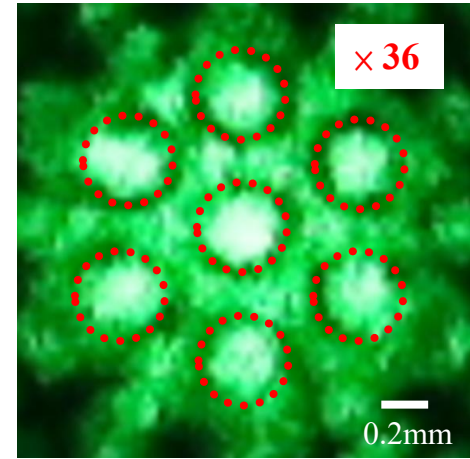
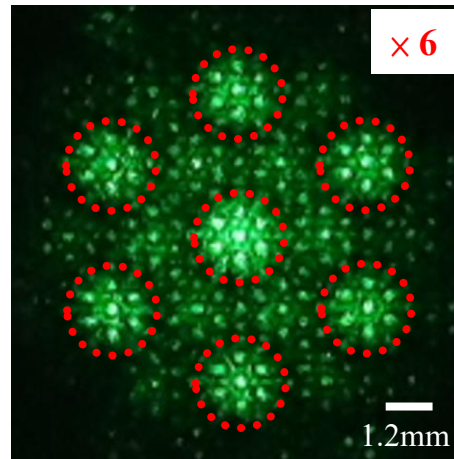
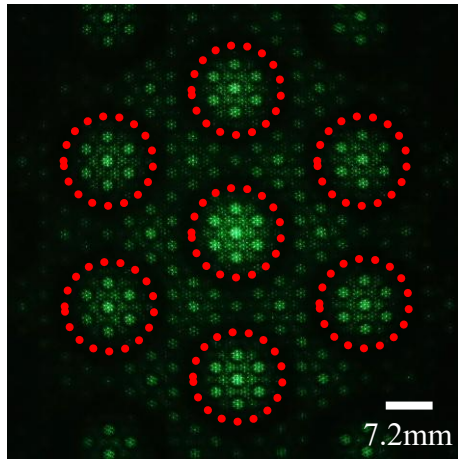
Comparison between double & triple layer



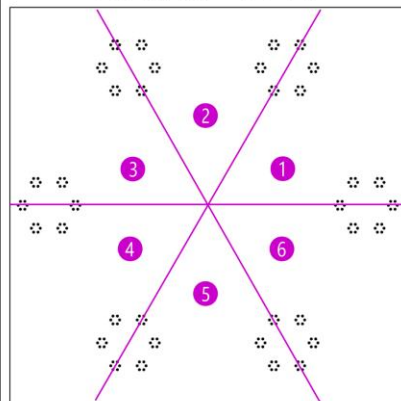
Discussion

Self-similar structure & rotational symmetry

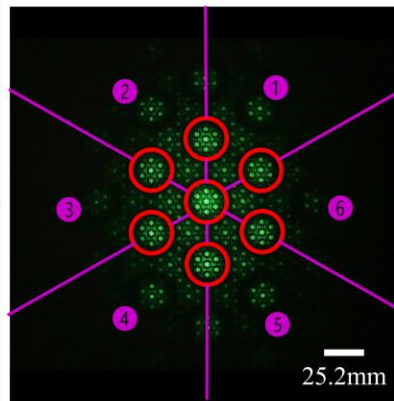
$$(Q_1, Q_2, Q_3) = (6, 6, 6)$$



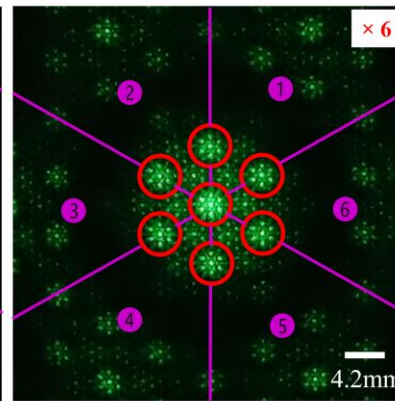
$$(Q_1, Q_2, Q_3) = (6, 6, 6)$$



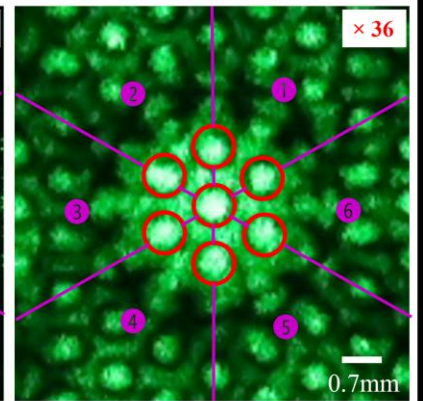
Far-felid



Far-felid zoom in

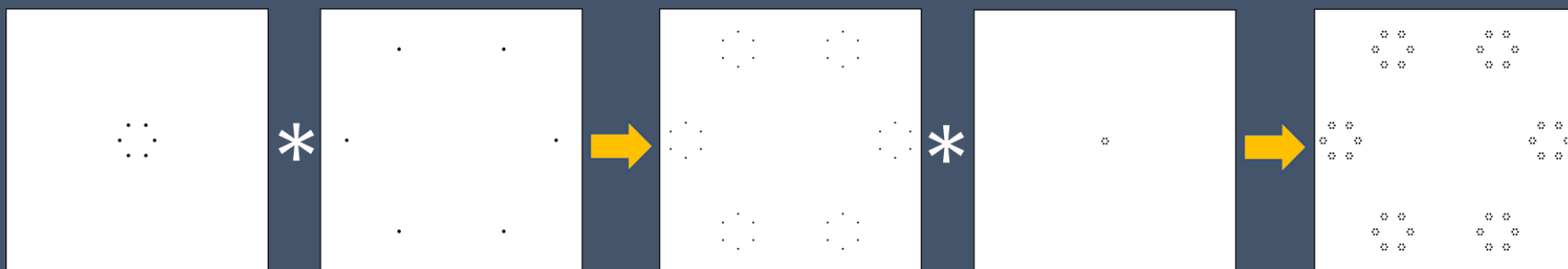


Far-felid zoom in zoom in

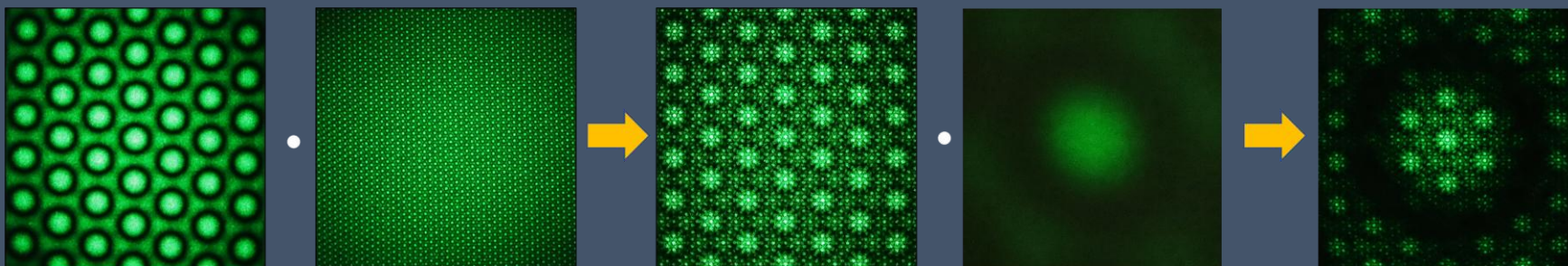


Discussion

Near field arrangement:



Corresponding far field interference pattern:



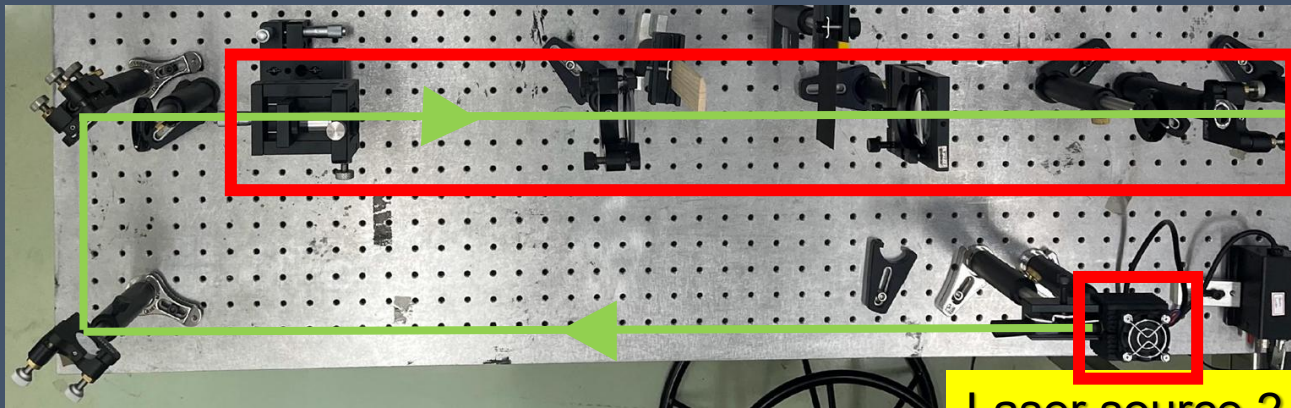
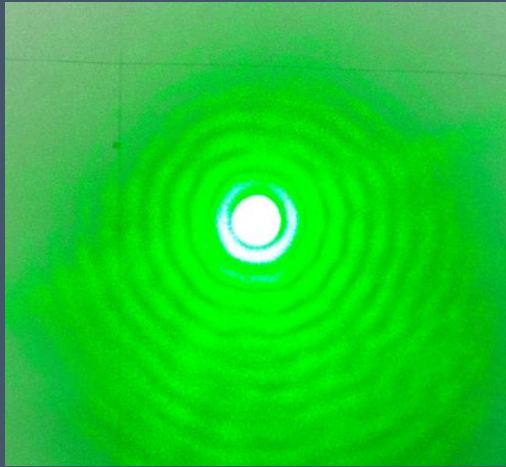
Conclusion

Conclusion

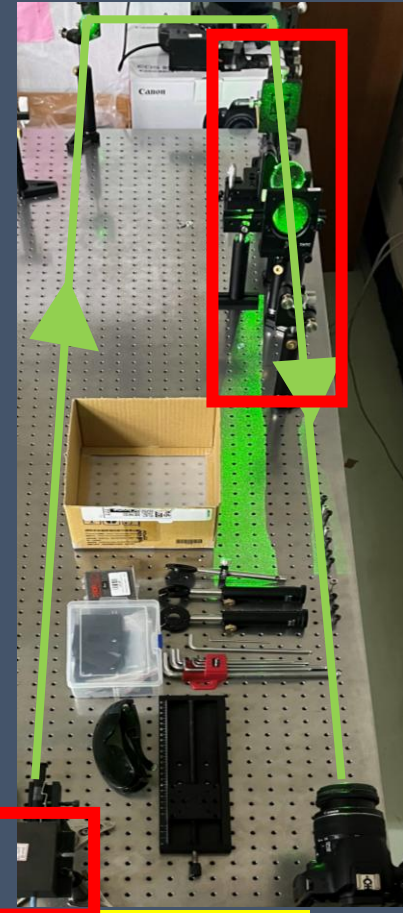
Tip-tilt error

Problem I met:

1. Spatial filter coupling with laser source.
2. Setting just right distance between each lens without tip-tilt error.



Laser source 2



Laser source 1



Conclusion

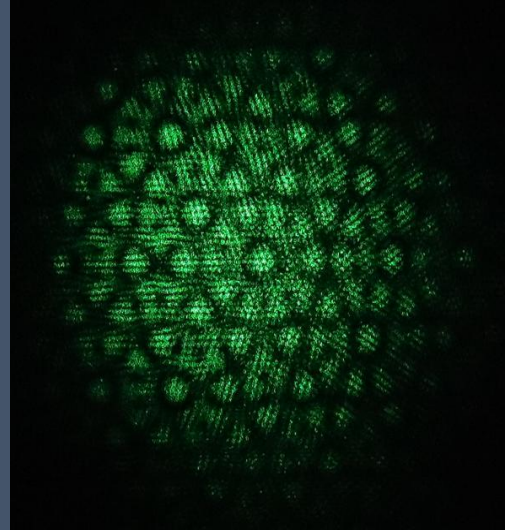
Problem I met:

3. Photomask quality.

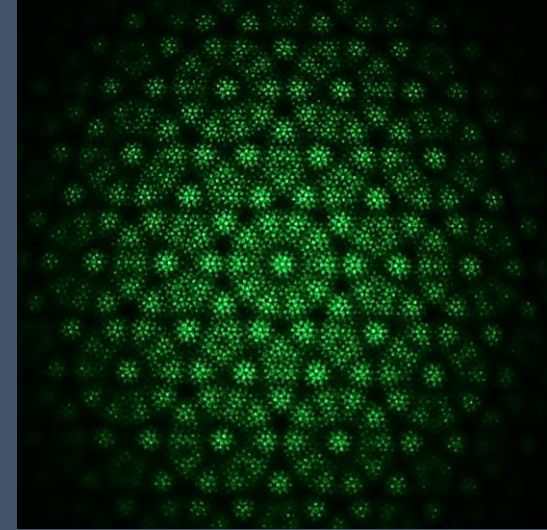
4. Reconstruction results.



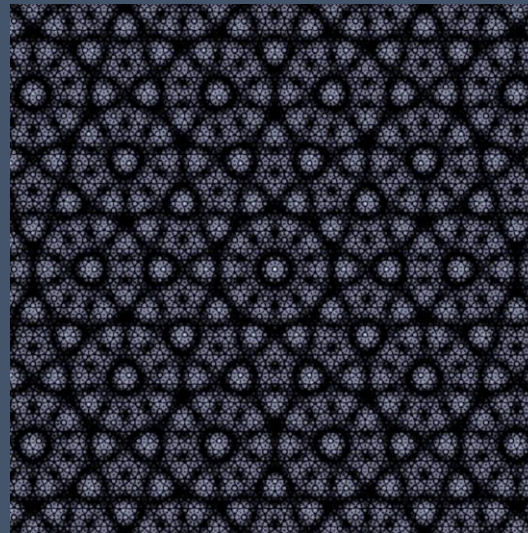
Exp. result by paper card



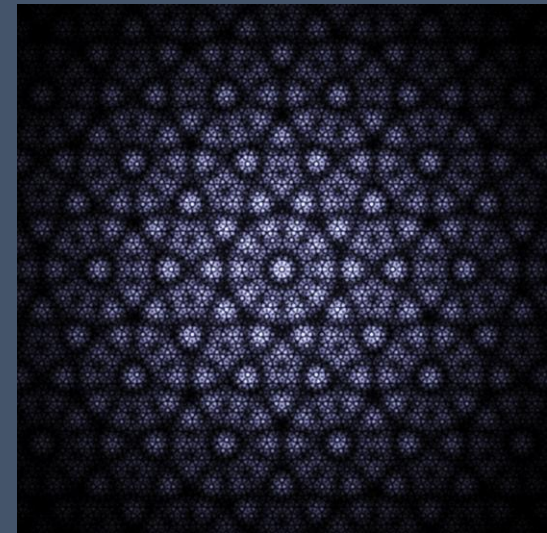
by aluminum sheet



Reconstruction by plan wave



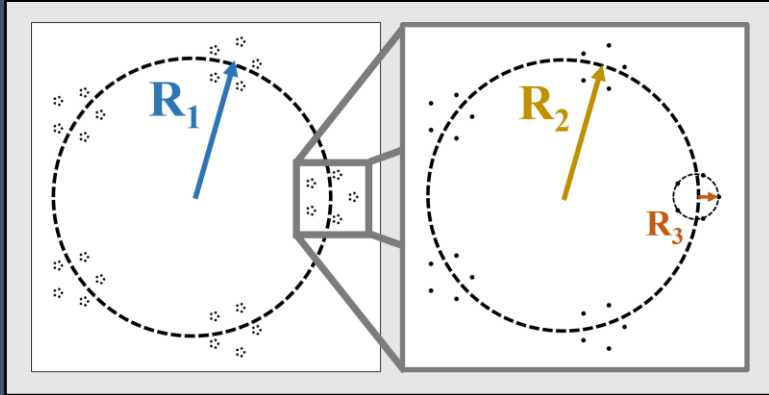
by Gaussian beam



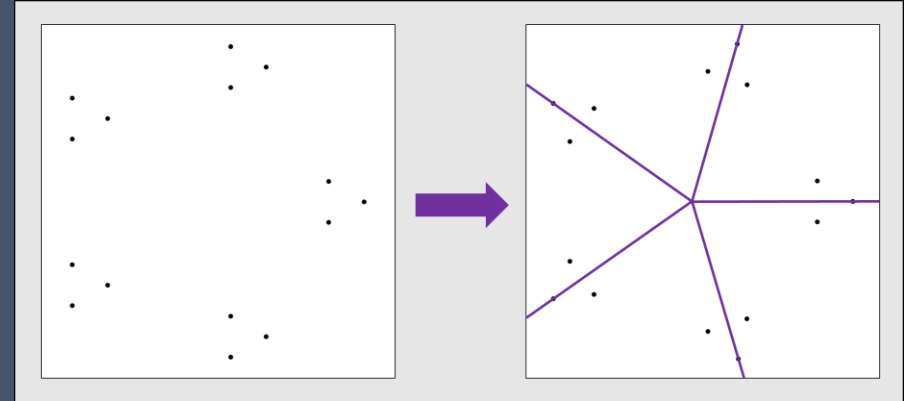
Future work

Future work

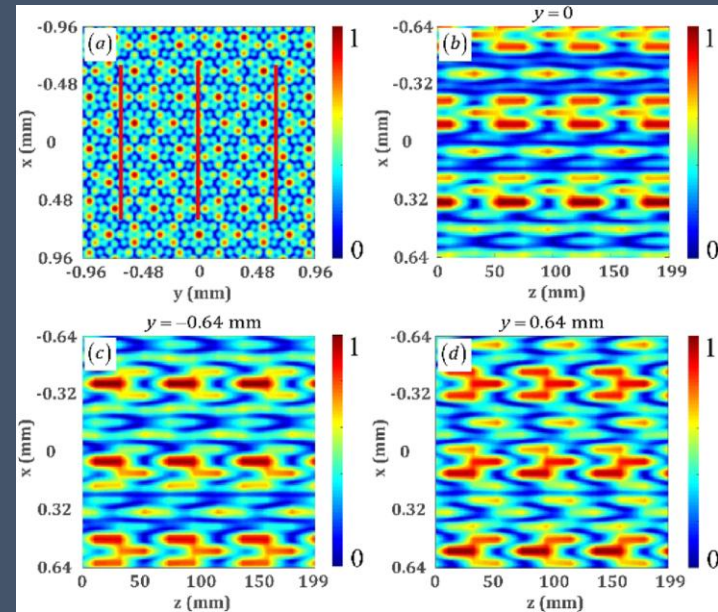
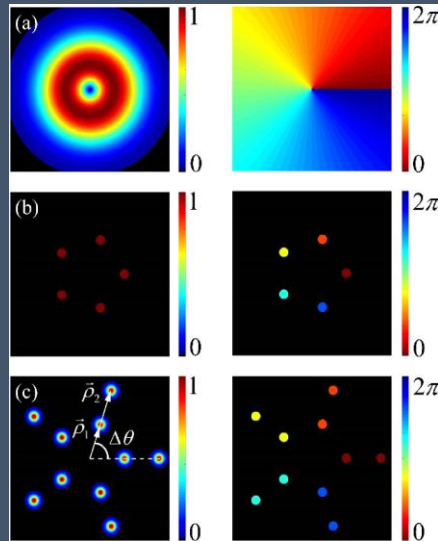
- Distribution distance coupling



- Aligning rotational symmetry axis

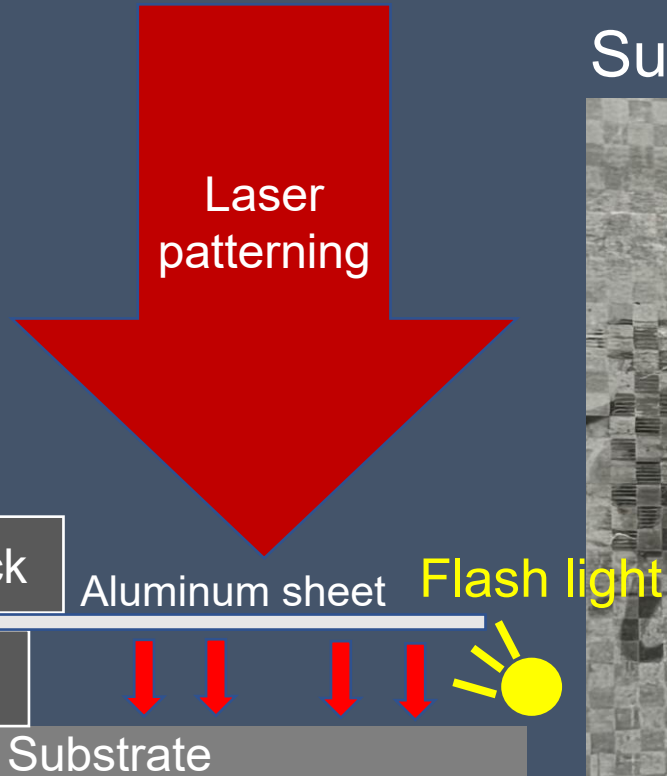


- Propagation evolution → Talbot effect

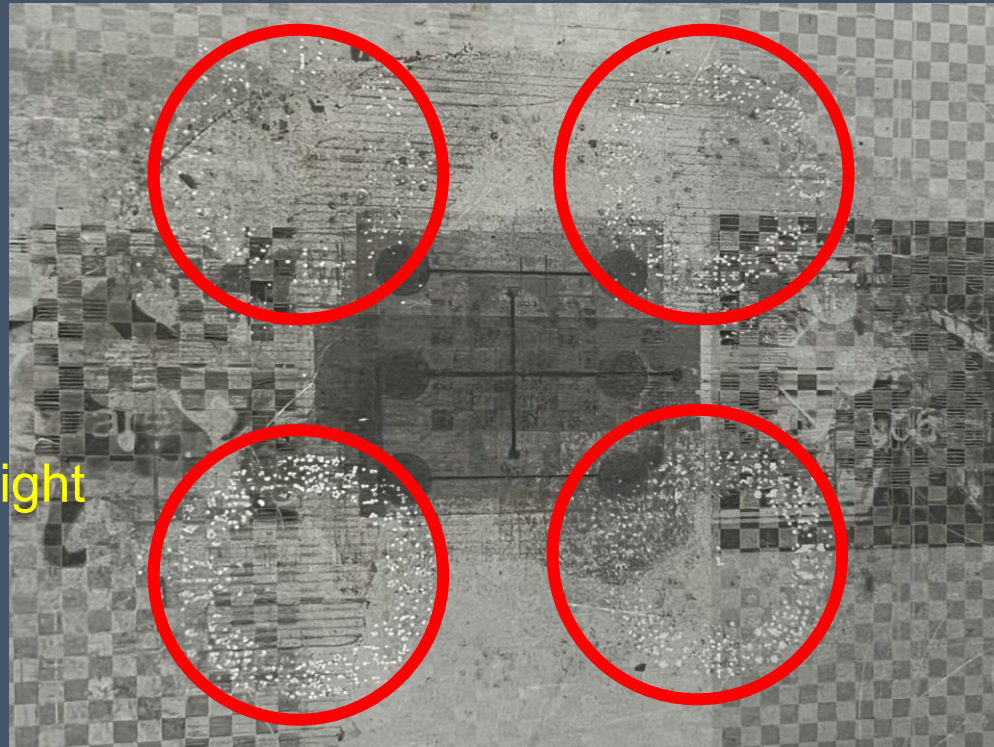


Apology

Apology



Substrate damage



Thanks for listening